rf-de-20m — Design Document

PCB-0007-A

hwde v1 pipeline

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1 Board and Run Metadata

Field	Value
board	rf-de-20m
part number	PCB-0007-A
workspace	`boards/PCB-0007-A_rf-de-20m`
phase	P9
gate status	3/5 gates pass (not passing: drc_routed, sim)
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2 Overview

3 Requirements

requirements.md - rf-de-20m

20 MHz Class E GaN RF power stage, 200 W into 50 ohm. Source: `brief/BRIEF.md` + `brief/RESEARCH-LEARNINGS.md`, owner-approved 2026-08-07.

Frozen inputs - not open for re-derivation. The topology, operating point and the two core parts below were decided by the owner after a full costed trade study. They are recorded here as settled requirements, not as proposals. Class DE was evaluated and REJECTED (no LCSC-stocked gate driver can switch a 20 MHz half-bridge). Do not reopen.

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1. Function

A bench RF power amplifier stage: a single-ended Class E inverter that converts a 40 V DC bus into 200 W of 20 MHz RF power delivered to a 50 ohm load. An externally generated 20 MHz PWM drive signal arrives on an SMA input, is buffered by a low-side GaN gate driver, and switches a single ground-referenced eGaN FET. The FET's drain feeds a shunt capacitance plus a series resonant tank and an L-match that transforms the 4.614 ohm Class E load line up to 50 ohm at the output SMA. Scope is the POWER STAGE ONLY. There is no MCU, no on-board PWM generation, no telemetry, and no protection logic on this board.

2. Interfaces

#	Interface	Direction	Standard / detail	Connector
I1	RF drive input	in	20 MHz logic-level PWM from an external signal generator; 50 ohm terminated at the board. Feeds the LMG1020 input. Adjustable duty cycle expected (~50% nominal for Class E).	SMA (stated in brief)
I2	RF power output	out	20 MHz, 200 W CW into 50 ohm. Controlled-impedance 50 ohm trace from the L-match to the connector.	SMA (stated in brief) - see Q10
I3	DC bus input	in	40 V DC, >=6 A.	NOT STATED - see Q5
I4	5 V housekeeping	in or internal	5 V rail for the LMG1020. Source not yet decided: onboard 40 V-input buck vs external 5 V feed.	see Q9

Notes carried from the brief that bind the interfaces:

- The 50 ohm output trace must be controlled-impedance all the way to the output connector.
- MUST-RESOLVE at design time: verify the LMG1020 input accepts a 50 ohm-terminated logic-level SMA signal directly; add bias or a fast buffer if it does not.
- No other external interfaces exist. No status LEDs, no buttons, no debug/telemetry header are required by the brief. ASSUMED: none wanted (scope is deliberately bare bones).

3. Power

Main bus (frozen):

Param	Value	Status
Bus Vdd	40 V DC	frozen
Bus current, DC average	~5.8 A at 85% efficiency	frozen
Supply requirement	>=6 A capable at 40 V	frozen
Output power	200 W into 50 ohm	frozen
Class E load line R_opt	4.614 ohm	frozen
Vds peak	142.5 V (3.562 x Vdd) -> 1.4x margin on the 200 V EPC2019	frozen
Ids peak	~16 A	frozen
Tank current	6.6 A rms (equals load current - loaded Q sets VOLTAGE across L/C, not current)	frozen
Expected efficiency	80-85% (NOT 86%)	frozen
Total dissipation	35-50 W, most of it in the passives, not the semiconductor	frozen

Housekeeping rail:

- 5 V for the LMG1020 gate driver. Must be a buck, NOT an LDO: gate-charge draw is Qg*f ~ 0.1 A, so a 40 V -> 5 V LDO would burn (40-5)*0.1 = 3.5 W.

- GUESS: 5 V rail budget 150-250 mA (0.1 A gate charge + driver quiescent + margin). To be confirmed against the actual EPC2019 Qg at the chosen drive voltage during design.
- If an onboard buck is used it must be LCSC-stocked and rated for a 40 V input (see Q9).

Frozen part selections:

- Switch: EPC2019 eGaN FET, 200 V / 36 mohm / Coss 110 pF / Qoss 18 nC @ 100 V. LCSC C2836675, ~$2.17. 1.35 mm chip-scale LGA, bottom-cooled through its solder balls.
- Driver: LMG1020YFFR, 5 V, 7 A source / 5 A sink, 3 ns prop delay, 1 ns min pulse, 60 MHz. LCSC C6423790, ~$0.36, 0.8 x 1.2 mm WCSP. NB: LCSC brands it "Tokmas", not TI - flag for an authenticity check on receipt.

Frozen output network (Class E, Q_L = 5):

Drain –+– C_shunt –GND C_shunt 317 pF TOTAL (Coss 110 pF + ~200 pF external C0G)

|

+– L_s – C_s –+– L_m –+– SMA out (50 ohm)

|

C_m – GND

L_s 184 nH | C_s ~447 pF | L_m ~115 nH | C_m ~500 pF (match Q 3.14, 4.614 -> 50 ohm)

Binding design constraints on that network (from the brief's MUST-RESOLVE list):

- The output inductor is the hottest component on the board (~7 W at Q=150), hotter than the GaN. Size L and C by DISSIPATION and Q, not by current rating. If no LCSC RF inductor carries 6.6 A rms at high Q: parallel several (splits current AND ESR), or use a PCB air-core spiral (JLC-native, free in BOM, costs board area).
- Parallel several C0G caps for C_s and C_shunt to split ripple current. Tank L/C see ~215 V peak -> specify >=250 V, ideally 500 V C0G. No X7R anywhere in the tank.
- The drain-feed RF choke must be a real RF choke with self-resonance well above 20 MHz.
- Decoupling: multiple small-case (0402/0603) C0G right at drain/source. Bulk electrolytic is useless at 20 MHz and exists only to support the 5.8 A DC feed.

4. Environment

NOT STATED in the brief. What is known:

- Bench instrument, no enclosure (explicitly out of scope). Ingress protection: none required.
- Vibration: none - stationary bench use. ASSUMED.
- Ambient temperature: not stated - see Q7.
- Cooling: 35-50 W of dissipation must leave the board. The EPC2019 is bottom-cooled through its solder balls, so its thermal path is vias straight into plane copper. Whether that copper is then coupled to a heatsink and whether forced air is present is NOT STATED - see Q6.
- Duty cycle (continuous 200 W vs pulsed/keyed) is NOT STATED and materially changes the thermal design - see Q8.

5. Size & mounting

NOT STATED in the brief - no outline, no mounting holes, no height limit given. See Q3 and Q4.
Constraints that are already fixed and will bound whatever outline is chosen:

- 4-layer stackup is MANDATORY (non-negotiable at 20 MHz). L2 must be an unbroken GND plane directly under the power loop.
- Board area is a design variable, not just a cost item: the fallback for the output inductor is a PCB air-core spiral, which trades board area for BOM cost and sourcing risk. A tight HARD outline cap could remove that fallback.
- Copper pours for the tank and power loop must be wide (skin depth in Cu at 20 MHz is 14.6 um; 1 oz = 35 um is only ˜2.4 skin depths), so conductor area cannot be shrunk to DC-current sizing.

WARNING: any outline cap answered as HARD binds permanently at P5 board_init and cannot be relaxed later without restarting layout. Answer HARD only if it is a real physical constraint.

6. Quantity & budget

NOT STATED in the brief - see Q1 and Q2.
The only budget guidance given is directional: "minimise spend by reducing SCOPE, never by reducing quality." Known BOM anchors: EPC2019 ˜$2.17, LMG1020YFFR ˜$0.36.

7. Assembly

- **100% JLCPCB PCBA. Zero off-catalog parts.** Owner ruled on this explicitly and it is the reason the topology is Class E rather than the originally requested Class DE. Every part must be LCSC/JLC stocked. This is a HARD constraint.
- No hand-soldered parts, no wound magnetics, no THT-only parts. (This is what rejected the gate drive transformer, the push-pull output transformer, and Mini-Circuits/Coilcraft RF transformers.)
- 4-layer PCB.
- Assembly sides: NOT STATED. ASSUMED single-sided (top) assembly, because the EPC2019 is bottom-cooled into plane copper and a clear bottom side is useful for heatsinking - confirm via Q6.
- The two connectors (I1, I2) and the DC input (I3) must be JLC-assemblable parts, which constrains the connector choices in Q5 and Q10.

8. Compliance / safety flags

All three of the following APPLY to this board.

1. **>30 V present (40 V bus, 142.5 V peak switch node).** The DC bus is 40 V and the drain node swings to a designed 142.5 V peak, with real ringing on top of that - the 1.4x derate against the EPC2019's 200 V rating is the entire margin. Consequences: creepage/clearance on the drain net and tank must be sized for >=215 V peak (tank L/C see ˜215 V pk); the 1.4x derate is protected by POWER LOOP AREA, not trace width, so layout is a safety item here, not just performance. Exposed high-voltage nodes on an unenclosed bench board are a shock/arc hazard.
2. **High current (>3 A): ˜5.8 A DC input, ˜16 A peak switch current, 6.6 A rms tank current.** Conductor sizing must account for skin effect at 20 MHz (14.6 um skin depth), not DC ampacity. The output inductor dissipates ˜7 W and is the hottest component on the board - a burn hazard on an open bench board and a fire/derating risk if the chosen part is sized by current

rating instead of dissipation. 3. **RF transmit: 200 W at 20 MHz.** This is a high-power HF RF source in a band adjacent to the 13.56 MHz and 27.12 MHz ISM allocations; 20 MHz itself is NOT an ISM allocation. EMC implications: a Class E stage is a hard-switched square-wave source, so harmonics at 40, 60, 80 MHz+ are strong and the only suppression is the output tank/match. Unenclosed, unshielded operation at 200 W will radiate. This board is for operation into a matched dummy load or a properly licensed/shielded setup only - it is not a certifiable transmitter and must not be connected to an antenna without an appropriate licence and additional filtering. 4. **Load-sensitivity hazard (consequence of the frozen topology).** Class E is narrowband and load-sensitive; it does not maintain ZVS across load pull. With protection logic explicitly out of scope, an open, shorted, or badly mismatched output at 200 W can drive Vds past the 1.4x margin and destroy the FET. This needs an explicit owner acknowledgement - see Q11.

Not applicable: no mains voltage anywhere on this board (external 40 V bench supply); no battery of any chemistry; no motors.

9. Open questions

Answer these and the design can proceed. Defaults are offered where a sensible one exists - if a default is fine, just say "default".

1. **How many boards do you want built?** (Default: 5 - JLCPCB's typical minimum-cost quantity for a 4-layer assembled board.)

2. **What is your target cost per board, assembled?** A rough ceiling is enough (e.g. "under $60 each"). This mainly decides whether the output inductor is bought (several paralleled SMD parts) or etched as a free PCB spiral that costs board area. (Default: no explicit ceiling - optimise for quality within a sensible bench-instrument cost.)

3. **Is there a maximum board size?** If yes, give width x height in mm, and say whether that limit is **HARD** (a real physical constraint, e.g. it must fit an existing fixture - this binds permanently and cannot be changed later) or **soft** (a preference). (Default: soft guidance of about 100 x 80 mm, no hard cap.)

4. **Do you want mounting holes?** If yes: how many, what screw size, and are their positions fixed by something existing? (Default: 4x M3 holes, one near each corner, positions chosen by the layout.)

5. **How should the 40 V DC bus connect to the board?** It must carry ˜6 A continuously. Options: (a) a screw terminal block, (b) a high-current 2-pin header, (c) solder lugs/pads for ring-terminal wires, (d) a banana-jack pair. Also: do you already have a 40 V supply that can deliver 6 A, and what does its output cable terminate in? (Default: (a) a 2-position screw terminal rated >=10 A.)

6. **How will the board be cooled?** It dissipates 35-50 W. Options: (a) bare board on the bench with no heatsink (this will get very hot and may not be survivable at continuous 200 W), (b) a heatsink bolted to the bottom of the board under the GaN FET, (c) a fan blowing across the top of the board, (d) both (b) and (c). Your answer decides whether the bottom copper must be left clear and flat, so it also fixes whether assembly is single- or double-sided. (Default: (d) - design for a bottom-side heatsink plus forced air, since that is the only answer that makes continuous 200 W comfortable.)

7. **What ambient (room) temperature should the board be designed to survive?** (Default: 25 C typical bench, designed with margin to 40 C.)

8. **Will this run at 200 W continuously, or in short bursts?** If bursts: roughly how long is a burst and how often (e.g. "10 seconds on, a minute off")? This changes the thermal design substantially - a pulsed duty cycle can make the hot output inductor a non-issue. (Default: continuous 200 W - the worst case, and the safe assumption.)

9. **Where should the 5 V driver supply come from?** Options: (a) an onboard step-down converter running off the 40 V bus, so the board needs only one power connection; (b) a separate 5 V input you feed from a bench supply or USB brick, which is one more cable but removes a part

and removes 40 V switching noise near the gate drive. (Default: (a) onboard buck - one cable is worth more on a bench instrument.)

10. **Confirm SMA for the 200 W RF output.** The brief says SMA and SMA is electrically fine at 20 MHz, but SMA is a small connector for 200 W and repeated mating at that power is a known wear point; a BNC or N-type is more robust. (Default: keep SMA as briefed.)

11. **SAFETY - please confirm explicitly:** this board has NO protection of any kind (no VSWR detection, no overcurrent, no overvoltage, no thermal shutdown - all were placed out of scope). Class E is load-sensitive, so running it into an open, a short, or a badly mismatched load at 200 W can push the switch node past its voltage margin and destroy the FET, and the RF output is a burn/RF-exposure hazard at 200 W. Do you confirm this will only ever be operated into a proper 50 ohm dummy load or matched load, on a bench, by you? (No default - this must be answered.)

12. **What does your signal generator actually put out on the drive SMA?** Specifically: peak-to- peak amplitude, whether it can drive a 50 ohm load, and whether its duty cycle is adjustable at 20 MHz. This decides whether the LMG1020 can be driven directly or needs a bias network or a fast buffer in front of it. (Default: assume a 50 ohm-capable generator putting out ~5 Vpp into 50 ohm with adjustable duty cycle, and design the input to tolerate 3.3-5 Vpp.)

—

10. ANSWERS - all 12 questions closed by the owner 2026-08-07 (P0 batch)

These are settled. Downstream phases treat them as requirements, not proposals.

```
| # | Answer |
|---|---|
| 1 | Build quantity **5** (default taken) |
| 2 | **No hard cost ceiling**; minimise spend by reducing scope, never quality
(default taken) |
| 3 | Outline **~100 x 80 mm, SOFT - no hard cap** (default taken).
Deliberately NOT hard-capped so the PCB air-core spiral inductor fallback stays
reachable |
| 4 | **4x M3** mounting holes at corners (default taken) |
| 5 | 40 V bus via **2-position screw terminal, >=10 A** (default taken) |
| 6 | **OWNER: continuous 200 W + bottom-side heatsink + forced air.** Bottom
copper stays clear and flat as a heatsink face -> **single-sided (top)
assembly** |
| 7 | Ambient **25 C bench, margin to 40 C** (default taken) |
| 8 | **OWNER: continuous 200 W.** Worst case; thermal drives the layout |
| 9 | **5 V from an onboard buck** off the 40 V bus - one cable (default taken)
|
| 10 | **OWNER: keep SMA.** 100 Vrms / 2 Arms at 20 MHz is within SMA HF limits;
N-type would break the 100% JLC PCBA rule |
| 11 | **OWNER SAFETY ACK: confirmed.** No VSWR / OCP / OVP / thermal protection
of any kind.  Operated into a proper 50 ohm matched or dummy load ONLY. Owner
accepts that an open, short or bad mismatch at 200 W will likely destroy the
FET |
| 12 | **OWNER: ~5 Vpp into 50 ohm, duty adjustable.** -> 50 ohm termination
direct into the LMG1020 input, **NO buffer stage**.  Closes brief MUST-RESOLVE
#4 |
```

Consequences that bind later phases

- **Single-sided top assembly.** Bottom layer is reserved as a flat heatsink mounting face; no bottom-side components. Thermal via array under the EPC2019 down to bottom copper.
- **Thermal is a first-class layout driver**, not an afterthought: 35-50 W continuous, and the output tank inductor (~7 W) runs hotter than the GaN.

- **No protection parts** are to be added by any later phase. If a reviewer flags the missing protection, the response is "waived, owner-acknowledged at P0" - not a new part.
- **Outline stays soft** through P5 `board_init`; do not pass a hard `--outline` cap that would foreclose a PCB spiral inductor.

4 Architecture

Decisions of record (state.json)

Phase	What	Why
P0	Thermal: continuous 200 W duty, bottom-side heatsink pad + thermal via array under EPC2019, forced air assumed; single-sided (top) assembly so bottom is a heatsink mounting face	Owner answer at P0 batch. 35-50 W loss with output inductor ~7 W hotter than the GaN; worst-case design chosen
P0	SAFETY ACK: no VSWR/OCP/OVP/thermal protection; matched or dummy 50 ohm load ONLY. Open/short output at 200 W will likely destroy the FET	Owner explicitly acknowledged. Class E loses ZVS under load pull; drain exceeds the 142.5 V design peak. Protection stays out of scope per power-stage-only brief
P0	RF output connector = SMA (edge/PCB mount, 50 ohm, LCSC-stocked)	200 W into 50 ohm at 20 MHz = 100 Vrms / 2 Arms, within SMA HF limits. N-type would break the 100% JLC PCBA rule
P0	Drive input: 50 ohm termination direct to LMG1020 input, NO buffer stage	Owner gen supplies ~5 Vpp into 50 ohm with adjustable duty. Closes brief MUST-RESOLVE #4; fewest parts and lowest added delay
P0	Defaults taken (owner did not specify): qty 5; no hard cost ceiling; outline ~100x80 mm SOFT (no hard cap); 4x M3 corner holes; 40 V bus via 2-pos screw terminal >=10 A; ambient 25 C bench w/ margin to 40 C; 5 V rail from onboard buck off the 40 V bus	Routine judgement calls. Outline deliberately left SOFT so the PCB air-core spiral inductor fallback stays available for the 6.6 A rms tank
P1	CORRECTED EPC2019 specs from datasheet: Coss(er) 156 pF (not 110), package 2.77x0.95mm solder bars (not 1.35mm LGA), Rds(on) 22 typ/42 max (not 36), Qg 2.4/2.9 nC	Brief figures came from a web summary, not the datasheet. Two P1 agents independently verified. External C_shunt drops to ~161 pF
P1	L_s and L_m to be etched as PCB air-core spirals, NOT catalog inductors	No LCSC part closes it: power inductors give Q 20-40 at 20 MHz (25-50 W); RF inductors rated 120-140 mA vs 6.6 A needed. Paralleling does NOT reduce loss (Q_total = Q_each). Spiral gives Q~130-150 at zero BOM cost
P1	Two-zone floorplan mandatory: heatsink zone (Q1/L1/bulk/driver) vs magnetics zone (L_s/L_m/tank caps) with L2 plane keepout under spirals only	A metal heatsink or solid ground plane under a PCB spiral is a shorted turn. L2 stays unbroken under the power loop; cut out ONLY under spirals

Phase	What	Why
P2	TWO EPC2019 in parallel (overrides the frozen single-FET brief)	H1 owner approval. Only config where thermal closes: Tj 103 C vs 138 C. 2x Coss(er)=312 pF meets the 317 pF shunt need, deleting the external cap. 2-FET layout can build with one populated; reverse is impossible
P2	Drive input is unipolar 0/+5V (generator TTL/CMOS mode); DC-coupled, no level shift, no AC coupling	H1 owner answer. LMG1020 input abs max -0.3 V forbids bipolar swing; AC coupling would destroy duty-cycle control that Class E depends on
P2	Controlled-impedance 50 ohm output trace RELAXED to a wide low-loss pour	H1 owner approval. At 20 MHz a 15 mm run is lambda/550, perturbs load <2%; the controlled-Z 0.4332 mm geometry would run 50-75 C at 2 A rms
P2	Outline 100x80 mm, P6 PRE-AUTHORISED to grow to 120x80 if a spiral cannot reach area. No hard –outline cap at P5	H1 owner approval. Spiral Q drives the dominant 16 W magnetics loss; ˜+$3-6/board is worth protecting it
P3	EPC2019 (C2836675) is OUT OF STOCK at LCSC (stock=0, price up 2.17->3.93). Design CONTINUES; order held pending restock. Re-verify at P10	Verified twice independently. No in-stock GaN substitute exists: 650V parts have Coss 239-760pF and 110-600mohm (8-11W conduction); EPC2051/2036 are 100-120V forcing a 25-33V bus. Stock-out does not change footprint/pinout/schematic, and it was in stock this morning, so design work is not wasted
P3	LMG1020YFFR primary = Tokmas C6423790; alternate = genuine TI-branded C506817 (in stock)	Part-sourcer found a real TI listing, resolving the architecture's authenticity flag
P3	RF choke L201 = 2x FXL0630-R47-M (470nH, 4.1mohm, 20A) in SERIES	No single LCSC part meets L/SRF/DCR/Isat together (SRF is essentially never published for adequately-rated parts). Uses the architecture's pre-authorised escape
P2	AMENDED operating point (Sokal finite Q_L=5 coefficients): R_opt 4.13 ohm, C_shunt 403 pF, L_s 164 nH, C_s 518 pF, L_m 110 nH, C_m 530 pF, I_tank 6.96 A rms. Bus STAYS 40 V	Frozen brief used 1/(wR*5.4466), which is Sokal's Q_L->infinity constant, but the board runs Q_L=5. Loss 38.6 W / 83.8%, inside the frozen band
P2	External shunt trim cap POPULATED at 3x33 pF (˜87 pF), not DNP	EPC2019 Coss spreads 110-150 pF (+36%). If device Coss is the whole shunt, that spread lands on ZVS with no absorber. The external cap is the only trim element
P2	TWO EPC2019 now MANDATORY (not merely preferred): single FET reaches Tj 160 C with a hypothetical 0 C/W heatsink, above the 150 C abs max	Real Rds(on) 36/50 mohm (65/90 hot). Pair gives 114 C nominal / 133 C max-corner at theta_HS <= 0.7 C/W. Dissipation is flat N=1->2; the win is RthJB halving per die
P3	SMA re-sourced to C22418168 CONSMA001-SMD-G-T (TE/Linx), verified TRUE SMD from manufacturer drawing; 100% SMT preserved. Cost 0.39 -> 2.97 each	Previous BWSMA-KE-P001 was a THT vertical panel jack despite an SMD label. LCSC package/type fields are unreliable for mount style; verified via pad-layer inspection + customer drawing
P3	EPC2019 footprint: end columns are 0.45 mm c-c (corrected 0.46 -> 0.450). The 0.68-0.70 mm figure is the OUTER ENVELOPE and was NOT applied	Fixer extracted datasheet vector geometry and refuted the librarian report. Applying 0.68 would have shifted the GATE pad 0.11 mm off a 0.25 mm pad

Phase	What	Why
P5	Board 120x80 mm, 4-layer JLC04161H-1080B, 1oz outer, 4x M3 corner holes. rules_gen: 23 rules, netclasses Pwr_5p5mm/Pwr_8p4123mm/Pwr_11p8942mm, HV clearances per net (TANK_A 180V)	Owner pre-authorised growth to 120x80; L301 courtyard fit at 100x80 was 0.01 mm, not a real margin. /SW needs 11.89 mm and +40V 5.5 mm track width at 200 W
P5	Added explicit Description field to L301/L302 in tank.py to match the spiral footprints' descr	Stock Device:L Description is 'Inductor', which mismatched the custom footprint descr and failed board_init schematic parity (2 -> 0)
P6	SMA jacks are vertical top-mount; cables exit upward. Owner accepted at H3	Only genuinely surface-mount 50R SMA at LCSC without B.Cu lands. Edge-launch would need THT + a heatsink notch or an off-catalog part
P6	12 intra-EPC2019 clearance errors to be WAIVED at P8 (0.35 mm die bar gap vs the 0.8 mm aiee_hv_143v_SW IPC-2221 rule)	Die geometry on a manufacturer-qualified 200 V part; IPC-2221 creepage governs board conductors, not a qualified component's internal terminals. No alternative part exists. U101's 4 solder_mask_bridge errors are NOT waived - those are a pulled-footprint artefact and get fixed
P6	All constraints.json rects translated to ABSOLUTE coords (outline origin 6.635,39.335). New B.Cu-only GND bridge board-local [49,30,94,48]; zone A extended 48->51; zone C In2/B.Cu at x=92, In1 held to x=96	planes_gen/placelib consume rects verbatim as absolute. Board-local rects meant zone A poured only y 0-41 (no plane under the buck) and zone C landed on top of L301. B.Cu bridge chosen over In2: 1oz vs 0.5oz, and 1.554mm below F.Cu so a TANK_A crossing couples ~4.7pF not ~29pF
P6	Spiral keepout radius NOT shrunk (stays r=20.55). Spirals nudged in y only; corridor 5.01 -> 6.02 mm, centre separation 46.84 mm	Widening 6->10 mm buys only 33 mW and 1.4 nH (0.5% of the 274 nH tank) against an unanalysed magnetics change to the Q 388/325 spirals. Nudge improves SPIRAL-5 margin and mutual coupling simultaneously
P8	DERATE to 36 V bus / ~162 W (from 40 V / 200 W)	H4 owner ruling. theta_BS was assumed 1.5 C/W but is realistically 4.05-7.3 C/W (1.065 mm core, 18 vias), putting Tj 151-188 C nominal vs a 150 C abs max. Derate buys ~14 C; the four board fixes buy ~20 C more. Bus voltage is a bench knob, raisable once real Tj is measured
P8	Revert gate resistors 6R8 -> 4R7	Common-source inductance measured 0.157 nH, ~5x smaller than the estimate 6R8 was bought against. Recovers ~4 C of Tj at no real ring cost (-0.85 V vs a -4 V floor)
P8	All 8 verify/drc waiver classes APPROVED by owner at H4	Creepage set audited independently: all 21 verified inside the two die pad bboxes, /SW pour min distance to GND 0.8004 mm - nothing hiding. check_thermal confirmed heatsink-blind in source
P8	OWNER REPRIORITISATION 2026-08-08: bench target is 25-30 V bus / 100-150 W with big fans. Thermals and max power DEPRIORITISED. Priorities are now (1) ZVS, (2) noisy-loop reduction, (3) does it work at all	Owner ruling. At 30 V, Vds pk ~107 V on a 200 V part and dissipation falls ~Vdd^2, so the theta_BS blocker stops binding. ZVS is LINEAR in Vdd, so the tank/shunt/match values stay valid unchanged at the lower bus - the derate costs nothing in correctness

Phase	What	Why
P8	SUPERSEDES the H4 fix list: drop the heatsink clamping holes, extra thermal vias and full 162 W re-derivation. Keep the B.Cu mask land + fiducials (cheap fab hygiene). Add: RUN the SPICE benches, which have never executed	Owner deprioritised thermals. SIM-1 (ZVS) and SIM-4 (tank/match) are the arbiters of whether the board functions, and verify_all has no sim_run member so the gate passed with them unexecuted
P8	P8 FIX a3 - ZVS POPULATE: C308/C309 DNP, C320 populated 27pF (C_s 504 -> 419 pF), C203 DNP (C_shunt bank 83 -> 27 pF). Applied in kicad/gen/tank.py + stage.py; parts.json quantities follow (56pF line 18 -> 15 populated of 21, 27pF line 2 -> 3 of 6). No new part, no new land, no netlist change.	The P8 SPICE benches (first execution of this board's declared sims) proved the board missed ZVS as built: ~30 nH of un-imaged zone-B copper in series with the C_s bank pushed X_net/R from 1.283 to 2.00, Vds at turn-on 15.6 V and P_out 53 W of 113 W. Extra series L is cancelled by LESS series C. Measured after: X_net/R 1.285, Vds 1.36-1.41 V, dVds/dt -1.21 V/ns, P_out 113.8 W, Vds,pk 122.3 V; holds across the whole 25-35 nH stray band.
P8	verify-waivers.json written at reports/verify-waivers.json, 8 entries, all approved 'Ihsan S. 2026-08-08 (checkpoint H4)'. gate verify PASS 0 failing / 31 waived.	Owner approved all 8 waiver classes at H4 (state.json human.4). Entries carry the reasons from reports/verify-waivers.md verbatim in substance. NB gate.py's default sidecar path is <pcb-dir>/reports/ (kicad/reports/), so the gate MUST be run with an explicit –waivers or the sidecar is silently ignored.
P8	OPEN: vds_pk_40v_v = 172.5 V against the 155 V derate line is the ONE remaining sim error and was NOT waived or weakened. One populate cannot serve both 30 V and 40 V.	A populate trimmed at 30 V is under-shunted at 40 V because the charge-equivalent Coss falls as the swing grows. At the declared 25-30 V bench the peak is 98.2-122.2 V and at the H4 36 V ruling it is 152.1 V, both inside. Owner ruling needed: declare the bus ceiling at 36 V (bench setting, zero board change) or re-populate C203 at 56 pF and re-run bus_derating before ever running 40 V.

Sheet plan

rf-de-20m - hierarchical sheet plan

AMENDED 2026-08-07 (P2-A): part values and net currents re-derived at the corrected operating point (**R = 4.13 ohm**, from Sokal's Q_L = 5 coefficients) on the real EPC2019 datasheet. The sheet split, refdes blocks and net-naming contract are unchanged. See `blocks.md` s0 and `decisions.md` D11.

Three sheets under a root that contains nothing but the sheet symbols and the inter-sheet wiring. ~60 parts. **Deliberately not decomposed further** - this is one function (a Class E stage) and the only organising principle that earns its keep is the floorplan, so the sheets map 1:1 onto the zones in `stackup.md` s3.

```
| Sheet | Blocks (‘blocks.md‘) | Zone | Refdes block | ‘#PWR‘ base |
|---|---|---|---|---|
| **‘hk‘** | B1 bus entry + bulk, B2 +5 V buck | A | **100-199** | **100** |
| **‘stage‘** | B3 drive input, B4 gate driver, B5 FET pair, B6 drain feed + bus
decoupling | A | **200-299** | **200** |
| **‘tank‘** | B7 series tank, B8 L-match, B9 RF output | B and C | **300-399** |
**300** |
```

Refdes are unique across sheets by construction: each sheet owns a hundreds block for every prefix,

and each sheet has its own #PWR base.

—

1. Net-naming contract - BINDING ON P4

KiCad names a net from a hierarchical label alone as **‘/<sheet>/<LABEL>‘**. A net that lives entirely inside one sheet therefore comes out with the sheet prefix, and `constraints.json` uses those names. **A net whose name does not match is silently no-op’d by every P5-P8 consumer** - this exact trap cost the lumina-par run a P4 amendment.

Power nets are **bare global symbols**: `+40V`, `+5V`, `GND`.

Exactly one signal net crosses a sheet boundary: ‘/SW‘. For it to be named `/SW` and not `/stage/SW`, **P4 must place a root-sheet local label spelled `SW` on the inter-sheet wire.** Everything else is sheet-internal and is named with its sheet prefix on purpose.

Canonical net	Scope	Carries
‘+40V‘	global	**5.96 A** DC nominal, 7.0 A declared. Ring to 51 V
‘+5V‘	global	**99 mA** avg (real Qg 1.8 nC), ~1.6 A/gate peaks
‘GND‘	global	In1 + In2 + B.Cu plane stack, zones A and C
‘/SW‘	**‘stage‘ -> ‘tank‘, root label required**	drain node. **9.17 A rms, 17.1 A pk, 142.5 V pk**
‘/stage/DRIVE‘	‘stage‘	J201 -> R201/R202 -> U201 IN+
‘/stage/GATE_ON‘	‘stage‘	U201 OUTH -> the two turn-on legs
‘/stage/GATE_OFF‘	‘stage‘	U201 OUTL -> the two turn-off legs
‘/stage/GATE_Q1‘	‘stage‘	gate of Q201
‘/stage/GATE_Q2‘	‘stage‘	gate of Q202
‘/tank/TANK_A‘	‘tank‘	L301 / C_s junction. **6.96 A rms, 156 V pk - the highest node on the board**
‘/tank/TANK_B‘	‘tank‘	C_s / L302 junction. 6.96 A rms, 41 V pk
‘/tank/RFOUT‘	‘tank‘	L302 -> C_m -> J301. 6.96 A rms at the C_m node, 2.0 A into the load, 141 V pk

The OUTH/OUTL nets are named ‘_ON‘/‘_OFF‘, NOT ‘_H‘/‘_L‘, on purpose. `rules_gen.detect_diff_pairs` auto-pairs `high_speed` nets whose names end in _H/_L (as well as _P/_N, _DP/_DM, +/-) and would silently treat the turn-on and turn-off legs as a 100 ohm differential pair, emitting a `diff_pair_gap` rule and an inner-layer `disallow track` rule on the two most inductance-critical nets on the board. Do not rename them back.

Do not ”tidy” ‘/stage/` or `/tank/‘ into bare names. Promoting a sheet-internal net to a root-crossing name means the root sees wire + label + one sheet pin, which raises ERC `label_dangling`; buying the bare name would mean adding a part to the root sheet. The netlist is the authority.

—

2. hk (100-199, #PWR 100) - zone A

Ref	Part / class	Note
J101	2-pos screw terminal, 5.08 mm, >= 24 A / 250 V (KF128-5.08-2P class)	**The only THT part on the board.** Left edge, x < 5 mm, clear of the heatsink land (HS-3)
C101, C102	2x 100 uF / **63 V** SMD polymer can	bulk. SMD can, not leaded - no bottom-face solder
C103, C104	2x 2.2 uF / 100 V **X7R** 0805	mid tier, 100 kHz-5 MHz. **X7R is correct here and nowhere else on this board**
U101	**LM5017-class 100 V synchronous COT buck**, SOIC-8-EP	Vin abs max >= 63 V; no comp network, no catch diode
L101	15-47 uH shielded, >= 0.5 A	placeholder value; **P4 sizes against the chosen Fsw**
C105	Vin local, 100 V	

```
| C106, C107 | VCC decoupler, bootstrap | per the buck datasheet |
| C108, C109 | Vout bulk + HF on `+5V` | |
| R101, R102 | feedback divider | sets 5.0 V +/-4% |
| R103 | RON / on-time resistor | sets Fsw |
```

Placement rule: at the DC-input end, as far from the gate loop and the tank as zone A allows. Its 0.5-2 MHz switching noise is spectrally clear of 20 MHz but its switch node is still a dV/dt source next to a 1.4 V gate threshold.

—

3. `stage` (200-299, `#PWR` 200) - zone A

This sheet is the board. Everything on it lives in one ~15 x 15 mm cluster except the drive connector.

```
| Ref | Part / class | Note |
|---|---|---|
| **J201** | SMD edge-launch SMA, 50 ohm (BWSMA-KE-P001 class) | **SMD, not the
THT bulkhead jack** - a THT jack solders through the bottom face and breaks the
heatsink land |
| R201, R202 | **2x 100 R 0805 in parallel** = 50 ohm | ~0.125 W in the
termination; a single 0603 is over its rating.  **DC-coupled - see `blocks.md`
B3** |
| **U201** | **LMG1020YFFR**, 6-ball WCSP 0.8 x 1.2 mm | **Sits ON the mirror
axis, equidistant from both gate bars.** LCSC brands it "Tokmas" - authenticity
check on receipt |
| C201 | 100 nF 0402 X7R | VDD bypass, **< 0.5 mm from the VDD ball, via-in-pad
return** |
| C202 | 10 nF 0201 C0G | second VDD bypass, straddling the ball |
| R203, R204 | 2x 4.0 R 0603 in parallel | OUTH -> `/stage/GATE_Q1` |
| R205, R206 | 2x 4.0 R 0603 in parallel | OUTH -> `/stage/GATE_Q2` |
| R207, R208 | 2x 4.0 R 0603 in parallel | OUTL -> `/stage/GATE_Q1` |
| R209, R210 | 2x 4.0 R 0603 in parallel | OUTL -> `/stage/GATE_Q2` |
| **Q201, Q202** | **2x EPC2019** 200 V eGaN | **MIRRORED PAIR about the U201
axis.** Both populated by default |
| C203-C206 | 4x 33 pF / 1 kV C0G 1206, **0-133 pF in 33 pF steps** | **C_shunt
trim - POPULATE 3 (99 pF) at P2-A**, no longer DNP. The FET pair supplies 316
pF of the 403 pF shunt; this bank supplies the balance AND is the only absorber
of the part's 110-150 pF Coss spread.  **In the power loop, not on a stub** |
| **L201** | RF choke, **>= 0.82 uH, SRF >= 80 MHz, DCR <= 25 mohm, I_sat >=
12 A, >= 8 A rms** | Hardest part on the BOM. **Pre-authorised escape:  2x 0.47
uH in series.** Not authorised:  a smaller choke - see `blocks.md` B6 |
| C207-C210 | 4x 10 nF / 100 V C0G 0603 | bus HF bank, **within 3 mm of L201's
bus-side pad** |
| C211, C212 | 2x 1 nF / 100 V C0G 0603 | terminates the bank's self-resonance |
```

Four resistors per FET pair per polarity is not over-specification. Individual gate resistors are what damp the differential mode between the two gate loops; a shared resistor leaves the two gate nodes coupled through the driver output and free to oscillate. Two 0603s in parallel per leg rather than one 0805 keeps each part at **0.029 W** (the real Qg of 1.8 nC drops the pair's gate-loop energy to 0.36 W) and roughly halves the leg's parasitic inductance. See `blocks.md` s3.

The gate-loop budget is the acceptance criterion for this whole cluster, and P2-A TIGHTENED it: <= 0.48 nH per FET, matched to +/-0.1 nH. The datasheet gives Ciss 200 pF and Crss 0.7 pF, so C_GS ~ 199 pF, and EPC WP008 Eq.1 at R_G + R_src = 3.1 ohm gives 0.48 nH (the retracted Qg of 2.4 nC had implied ~350 pF and 0.84 nH). **This is the tightest layout spec on the board** - two parallel 0603s are ~0.15-0.2 nH, leaving ~0.3 nH for vias and interconnect. **Stated fallback: R_G = 3 ohm relaxes the budget to 0.84 nH** at ~+0.3 W turn-off loss and ~+1.6 C of Tj; take it consciously.

Match the loops geometrically, not the trace lengths electrically - FR4 is ˜6.7 ps/mm, so length matching is not the mechanism. The +/-0.1 nH exists to damp the differential mode and equalise static sharing; **skew itself is benign** in a soft-switched topology (at turn-on the drain is at ˜0 V, so an early device has nothing to hog).

Common source star. Both source bars return to **one** via cluster on the mirror axis, and the LMG1020's GND connects to the source plane **only at that star point** (TI LMG1020DS s10.1.1 - single-point ground). Separate source returns would make common-source inductance differential and imbalance the pair.

—

4. `tank` (300-399, `#PWR` 300) - zones B and C

```
| Ref | Part / class | Zone | Note |
|---|---|---|---|
| **L301** | **L_s 164 nH - etched PCB air-core spiral** | B | 2 turns, OD 30-34
mm, **˜2.5-3 mm trace** (wider than before - the lower L target at the same OD
needs it, and it buys Q), **>= 1430 mm^2 at Q 100**.  **No LCSC code.  'PCB
feature - do not place'.** SPIRAL-1..6 |
| C301-C309 | **9x 56 pF / 1 kV C0G 1206 = 504 pF** (target 518 pF **+/-5%**) |
B | C_s.  0.77 A rms and ˜105 mW each.  **No X7R.** The +/-5% is OPEN-12:  the
P1 fragment's C_series coefficient is refuted and SIM-4 is the arbiter - keep
it a parallel bank so it trims by depopulation |
| **L302** | **L_m 110 nH - etched PCB air-core spiral** | B | **>= 950 mm^2 at Q
100 - nearly as large as L301 despite being 67% of the inductance**, because
the constraint is dissipation area, not L (SPIRAL-3) |
| C310-C319 | **10x 1206 C0G 1 kV = 530 pF +/-3%** | C | C_m.  Carries **6.66 A
rms**, not the 2 A load current - 0.67 A each |
| **J301** | SMD edge-launch SMA, 50 ohm - **same part as J201** | C | right
edge.  100 Vrms / 2.0 A rms |
```

Why C_m sits in zone C and C_s does not. C_s is a *series* element and floats - it needs no ground reference and belongs beside the coils. C_m returns to GND and carries 6.66 A rms, so it must sit where the ground plane exists.

L301 and L302 are schematic symbols with custom footprints, not annotations. The footprint carries the spiral copper on F.Cu (and B.Cu under SPIRAL-2), a courtyard equal to the thermal footprint, and its two terminal pads. Copper that no tool knows about gets routed over, poured under and placed on top of.

—

5. Placement plan for P6

5.1 Placement groups (declared in `constraints.json.placement.groups`)

```
| Group | Anchor | Members | Purpose |
|---|---|---|---|
| 'switch' | **Q201** | Q202, U201, R203-R210, C201-C212, L201 | **The whole
thing.** Power loop + gate loop + choke + bus HF in one ˜15 x 15 mm cluster |
| 'drive_in' | J201 | R201, R202 | 50 ohm termination at the connector, not at
the driver |
| 'tank_ls' | L301 | C301-C309 | C_s bank adjacent to the L_s spiral |
| 'tank_lm' | L302 | C310-C319, J301 | L_m -> C_m -> SMA, in that order, shortest
possible |
| 'hk' | U101 | L101, C105-C109, R101-R103 | buck and its loop |
| 'bus_in' | J101 | C101-C104 | bulk at the terminal |
```

5.2 Edges

Ref	Edge	pos	Why
J101	**left**	0.5	DC entry. **x < 5 mm so its THT pins clear the heatsink land** (HS-3)
J201	**top**	0.2	drive input, in zone A near the driver
J301	**right**	0.5	RF output, in zone C. Must be adjacent to the C_m node

5.3 Parts that must be placed by explicit `place_edit` and LOCKED before the annealer runs The annealer optimises a cost function; none of the constraints below are expressible in it.

Ref	Why locking is mandatory
Q201, Q202	mirror symmetry about the U201 axis is the whole of ‘blocks.md‘ s4.1(c). An annealer will not produce a mirror pair
U201	must sit **on** the axis, equidistant from both gate bars
R203-R210	four matched legs; two arms must be geometric mirrors
L301, L302	zone B, centre-to-centre >= 38 mm, >= 14 mm clear of the heatsink land (SPIRAL-5, SPIRAL-6)
J301	zone C, adjacent to the C_m node, so ‘/tank/RFOUT‘ stays <= 15 mm
J101	x < 5 mm (HS-3)

After L301 and L302 are placed and locked, hand-add KiCad rule areas (keepouts) over both courtyards on all four layers. No pipeline check enforces "no other copper here" and `planes_gen` has no void support. Verify geometrically at P8.

5.4 Zone assignment for P6 (board-local, translate by `outline_bbox` first)

Zone	x	Sheets	Contents
A	0 - 48	‘hk‘, ‘stage‘	everything except the tank
B	48 - 88	‘tank‘	L301, L302, C301-C309. **Nothing else may be placed here**
C	88 - 100	‘tank‘	C310-C319, J301

Heatsink land on B.Cu: **[5, 10, 36, 70]**, declared as a back-side keepout.

—

6. Sheet-level P4 notes

1. **The C_shunt trim bank (C203-C206) must be in the netlist, and at P2-A it is POPULATED (3 of 4 sites, 99 pF), not DNP.** The FET pair supplies 316 pF of the 403 pF shunt; the bank supplies the balance and is the **only** mechanism that absorbs the part's 110-150 pF Coss spread. The fourth site stays unpopulated as headroom. All four have pads either way, so their clearance and area are accounted for at P6/P7 and the trim stays a populate change rather than a respin. 2. **‘/SW‘ needs a root-sheet local label** or it becomes `/stage/SW` and six `constraints.json` entries silently stop matching. 3. **Both spirals need schematic symbols.** Do not let them become a layout-only feature; their nets must be real. 4. **The drive input is DC-coupled.** Do not add a series blocking capacitor "for safety" - it destroys duty-cycle control (`blocks.md` B3). Record it on the schematic. 5. **No protection parts.** No TVS, no fuse, no clamp, no OCP, no OVP, no thermal shutdown. Owner-acknowledged at P0 Q11. If ERC or a reviewer wants one, the response is *"waived, owner-acknowledged at P0"*. 6. **L201's value is a floor, not a target.** >= 0.82 uH. If P3 cannot source it, the answer is 2x 0.47 uH in series - **not** a smaller single part. 7. **‘IN-‘ on U201 ties to GND.** No bias divider, no buffer. 8. **Silkscreen hazard markings**: zone B (two ~130 C copper structures, one on each face if SPIRAL-2 is adopted), `/tank/TANK_A` (**156 V pk**), `/SW` (142.5 V pk), and J301 (200 W RF - burn and RF-exposure hazard). This is an unenclosed bench board with no interlocks of any kind. 9. **Silkscreen the bus voltage as a**

RANGE, not a value: "40 V max, 34-40 V". P2-A makes the bus a deliberate bring-up and derating knob (`decisions.md` D1), and ZVS is Vdd-independent, so a 36 V bench setting is a valid operating point at ~160 W - not a fault.

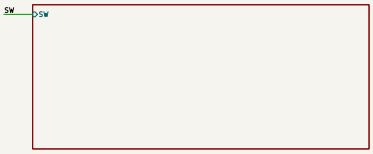
Full architecture narrative (not inlined; see the artifact index): `architecture/blocks.md`, `architecture/decisions.md`, `architecture/power_tree.md`, `architecture/stackup.md`.

5 Schematic

Gate `erc` (phase P4): **pass** after 1 attempt(s); last 2026-08-08T23:51:21, failing 0 of 0. The full schematic PDF follows.







ROOT SHEET — NO PARTS BY DESIGN (sheets.md §1).

The ONLY inter-sheet signal net is /SW (the drain node, 142.5 V pk, 9.17 A rms). Its bare /SW spelling comes from the root-local labels written by add_sheet, and six constraints.json entries depend on it. Both 'stage' and 'tank' expose it, which is what keeps the root label connected — a root label with only ONE sheet pin raises label_dangling in KiCad 10.

GND / +40V / +5V are global POWER SYMBOLS inside the sheets and are deliberately absent here. All three PWR_FLAGs live on 'hk'; a second flag anywhere else collides power_out <-> power_out.

Every other net is sheet-internal on purpose and carries its /<sheet>/ prefix. Do not 'tidy' them into bare names.

al-ee

Sheet: /

File: rf-de-20m.kicad_sch

Title: rf-de-20m: root — 20 MHz Class E GaN stage, 200 W into 50

Size: A3	Date: 2026-08-07	Rev: 1
KiCad E.D.A. 10.0.3		Id: 1/4


PWR_FLAG
GND
+40V
+5V
THIS SHEET OWNS THE PWR_FLAGS for GND, +40V and +5V. Every rail enters through passive glue (J101) or a passive inductor (L101). Do NOT add a second flag on another sheet PWR_FLAG is a power_out pin and two of them collide as power_out <-> power_out.
J101
KF128-5.08-2P 2pos 5.08mm 24A/250V
C101 100uF 63V alu electrolytic
C102 100uF 63V alu electrolytic
C103 2.2uF 100V X7R 1206
C104 2.2uF 100V X7R 1206
TYPE 3 RIPPLE INJECTION IS MANDATORY, NOT OPTIONAL. LM5017 is a COT regulator and needs >=25 mV of IN-PHASE ripple at FB (s7.3.1/s7.3.11). A 22 uF ceramic COUT gives almost none, and the regulator misfi-pulses. R104/C110/C111 synthesize the ramp from the SW node so the +5 V rail keeps low ripple — which matters, because +5V feeds a driver whose VDD abs max is 5.75 V. FB ripple 78.6 mV pk-pk at 40 V bus (3.1x the floor); R×C× is 3.1x inside Eq 7.
U101
GND
BTN
VIN
+40V
UVLO
RON
SW
BST
VCC
FB
PAD
BUCK_SW
GND
LM5017 100V sync COT buck
C105 100nF 100V X7R 0603
C106 2.2uF 50V X7R 0805
THE BUCK SWITCH NODE IS /hk/BUCK_SW. It is NOT called SW: /SW is the RF drain node and is named in six constraints.json entries.
UVLO (pin 3) IS TIED TO VIN. Datasheet s7.3.9 sanctions it explicitly: no bus-undervoltage threshold is specified anywhere in the architecture, so a divider would be an invented requirement. Abs max 100 V.
C101/C102 ARE POLARIZED: pin 1 = ANODE (+). SETTLED 2026-08-08 against the manufacturer drawing (JOHO RVT series, parts/C51953411.pdf p.2): the 'Positive' leader points at the CHAMFERED-corner end of the base, and the footprint's two chamfers are on the pad-1 side. The symbol's '+' at pin 1 is therefore CORRECT. (The usual 'cut corner = NO PROTECTION PARTS ON THIS BOARD, cathode' rule does NOT hold on this family: No TVS, fuse, NTC, OVP, OCP or thermal — the can's black top stripe sits on the shutdown. Owner-acknowledged at PS Q11. SQUARE-cornered end. Cross-checked against The <51 V turn-on ring is handled by LumEmax SMDGP/RVT, same conclusion.) VOLTAGE RATING (every >40V part >=63 V) The footprint still carries NO silk polarity and by MORE bulk — shrinking the bulk makes marker — PG MUST add a '+' outside the the overshoot WORSE (blocks.md R1). 10.46 mm body square (the base hides it) and an F.Fab body outline with a marker.
22uH 1A shielded L101
BUCK_SW
+5V
C107 BUCK_SW 10nF 100V X7R 0603
C108 +5V GND 22uF 16V X7R 1206
C109 GND 100nF 100V X7R 0603 +5V
R101 +5V FB 30.9k 0402 1%
R102 FB GND 10.0k 0402 1%
R103 RON +40V 100k 0805 1% 150V
R104 BUCK_SW RINJ 100k 0805 1% 150V
C110 +5V 1nF 100V C0G 0603 RINJ
C111 FB 100nF 100V X7R 0603 RINJ
al-ee
Sheet /hk/
File: hk.kicad_sch
Title: rf-de-20m: hk — 40 V bus entry, bulk, +5 V buck
Size: A2 Date: 2026-08-07 Rev: 1
KiCad E.D.A. 10.0.3 No: 2/4

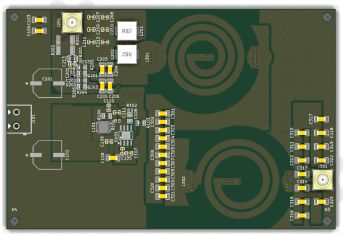

GND
+5V
+40V
+5V_DRV
PWR_FLAG
SW
J201
SMA Jack 50R SMD (CONSMA001)
FB201
+5V_DRV
DRAIN
R201
100R 2010 1% 750mW
R202
100R 2010 1% 750mW
U201
IN-
IN+
OUTH
OUTL
GND
DRIVE
GATE_OFF
LMG1020 5V GaN driver 7A/5A
C201
100nF 16V X7R 0402
C202
10nF 25V X7R 0201
C153
1uF 10V X7R 0603
GATE_G01
GATE_G02
487 0805 1% 250mW
Q201
Q202
SOURCE
DRAIN
EPC2019 200V eGaN
C203
C204
C205
C206
56pF 1kV C0G 1206
DNP
470nH 20A 4.1mohm
L201
L201_MID
L202
SW
C207
C208
C209
C210
10nF 100V X7R 0603
C211
1uF 100V C0G 0603
C212
1uF 100V C0G 0603
EPC2019 PIN 7 IS SUBSTRATE AND IS TIED TO SOURCE. The datasheet states it three times. It is NOT expressible in the footprint — this schematic is the only place it is enforced. All SOURCE pins (2/4/6/7) -> GND; both DRAIN pins (3/5) -> /SW.
OUTH AND OUTL ARE SEPARATE OUTPUTS. OUTH sources only and floats when idle: OUTL sinks only and is held low otherwise. NO steering diode is needed or wanted. TI'S ==2 OHM MANDATE IS AT THE PIN, AND TWO FETs SHARE EACH PIN, so each branch must be >= 2 x 2 = 4 ohm. ONE 487 0805 per leg per FET -> 4.7/2 = 2.35R at OUTH and at OUTL (2.33R at -15). peak drive 2.13 A. Do NOT parallel these back down. IN- IS TIED TO GND. DRIVE GOES TO IN+. Truth table: only (IN-=L, IN+=H) drives the gate high. IN- has an internal pull-UP to VDD, so floating it holds the stage OFF: IN+ has an internal pull-DOWN, so an open drive input fails SAFE. TI's ground-bounce alternative (drive IN-, IN+ to VDD) is NOT used: it inverts, and duty cycle is the primary ZVS trim.
THE DRIVE INPUT IS DC-COUPLED — RULING DB. Do not add a series blocking capacitor (for safety!) AC coupling's DC restore pins the average at the bias point and at D=0.4 both logic levels sit above the 1.4 V threshold, so the FETs never turn off. Generator MUST be unipolar 0 to +5 V — the input abs max is −0.3 V to VDD+0.3 V, so +/−2.5 V bipolar violates it (OPEN-1).
C203–C206 ARE 56 pF, NOT 33 pF. No 33 pF part exists on this BOM; parts.json maps these sites onto the same 56 pF 1 kV C0G part as the tank banks. Populate count re-solved: 2 x 56 = 112 pF external, giving 316 + 112 = 428 pF against a 403–449 pF requirement — mid-band. C205/C206 are DNP headroom; the bank EMPTIES for a max-Coss pair, which is the point of it existing.
GATE LOOPS: aim for <= 0.48 nH per FET, matched +/−0.1 nH, geometrically MIRRORED about the U201 axis. The CRITICAL-DAMPING budget is now 1.70 nH (EPC WPDOR Eq.1 at R = 4.7 + 0.4 + 0.75 = 5.85 ohm, C_GS 199 pF) − 3.6n looser than the 0.48 nH solved at the old 3.1 ohm, which is what pays for the 0805 bodies. Keep aiming at 0.48 nH. Do NOT length-match electrically — FETs is 6.7 ps/mm and skew is benign in a soft-switched topology; matching damps the differential mode and equalises sharing. L201 + L202 ARE ONE CHOKE, IN SERIES. 0.94 uH total — the architecture's own pre-authorised escape. A smaller single choke is NOT authorised: it breaks the Ideal Class E equations and RAISES the required shunt capacitance.
NO PROTECTION PARTS. No TVS on /SW, no gate clamp, no negative bias. Owner-acknowledged at PD Q15. The 1.40x derate on a 200 V part is protected by POWER LOOP AREA at P6/P7, FB201 IS A FILTER, NOT PROTECTION.
+5V IS SPLIT AT FB201: +5V_DRV feeds U201/U202 and the C153 only. That containment is the
the 72–180 mA of gate charge at 20 MHz must
return inside the local 1.1 nF, not across
Title: rf-de-20m: stage — drive, LMG1020, EPC2019 pair, C_shunt trim,
Sheet: /stage/
File: stage.kicad_sch
Size: A2
Date: 2026-08-07
Rev: 1
KiCad 10.0.3
Sh: 2/4


GND
GND
SW
C1W
SW
L301
164nH
TANK_A
TANK_B
L302
110nH
RFOUT
J201
GND
GND
GND
GND
RFOUT
SMA jack SIDE SMD (CONSMA001)
C301
TANK_A
TANK_B
56pF 1kV C0G 1206
C302
TANK_A
TANK_B
56pF 1kV C0G 1206
C303
TANK_A
TANK_B
56pF 1kV C0G 1206
C304
TANK_A
TANK_B
56pF 1kV C0G 1206
C305
TANK_A
TANK_B
56pF 1kV C0G 1206
C306
TANK_A
TANK_B
56pF 1kV C0G 1206
C307
TANK_A
TANK_B
56pF 1kV C0G 1206
C308
TANK_A
TANK_B
56pF 1kV C0G 1206
C309
TANK_A
TANK_B
56pF 1kV C0G 1206
C320
TANK_A
TANK_B
27pF 1kV C0G 1206
DNP
C321
TANK_A
TANK_B
27pF 1kV C0G 1206
DNP
C310
RFOUT
GND
56pF 1kV C0G 1206
C311
RFOUT
GND
56pF 1kV C0G 1206
C312
RFOUT
GND
56pF 1kV C0G 1206
C313
RFOUT
GND
56pF 1kV C0G 1206
C314
RFOUT
GND
56pF 1kV C0G 1206
C315
RFOUT
GND
56pF 1kV C0G 1206
C316
RFOUT
GND
56pF 1kV C0G 1206
C317
RFOUT
GND
56pF 1kV C0G 1206
C318
RFOUT
GND
56pF 1kV C0G 1206
C319
RFOUT
GND
27pF 1kV C0G 1206
C322
RFOUT
GND
27pF 1kV C0G 1206
DNP
C323
RFOUT
GND
27pF 1kV C0G 1206
DNP
L301 AND L302 ARE ETCHED PCB AIR-CORE SPIRALS.
They are real components with real nets and NO FOOTPRINT —
PB authors the copper. No LCSC code, excluded from the BOM.
SPIRAL-L: L ∝ r²/(8r + 11w), Q >= 120 at 20 MHz, copper area >= P/7 mW.mm⁻²
(L301 >= 1430 mm2 at Q 100, L302 >= 950 mm2 at Q 100).
SPIRAL-C: centre-to-centre >= 38 mm, and PB must compute the
residual mutual coupling k and fold it into the tank solve —
the two spirals are NOT independent components.
SPIRAL-G: no metal within 15 mm in plan view on either face.
A conductive plate under a spiral is a SHORTED TURN and no
copper cutout prevents it — that includes the heatsink,
brackets, standoffs and fasteners.
Zone B carries no plane on In1/In2/BL.Cu by construction, and a
al-ee
Sheet: /tank/
File: tank.kicad_sch
Title: rf-de-20m: tank — L_s/C_s series tank, L-match, RF output
Size: A2
Date: 2026-08-07
Rev: 1
KiCad E.D.A. 10.0.3
Sh: 2/4

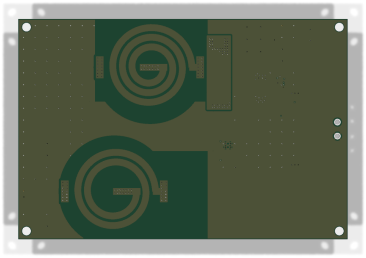
6 Layout

Gate `place`: *not recorded.*

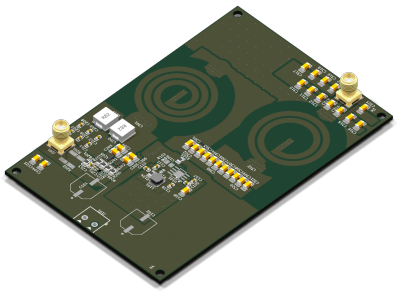
Gate `drc_routed` (phase P7): **fail** after 1 attempt(s); last 2026-08-08T23:51:20, failing 55 of 55.



`reports/renders/rf-de-20m_top.png`



`reports/renders/rf-de-20m_bottom.png`



`reports/renders/rf-de-20m_iso.png`

7 Verification

Gate `verify` (phase P8): **pass** after 1 attempt(s); last 2026-08-08T23:51:20, failing 0 of 189. *reports/verify_all.json not available.*

Design review of record

rf-de-20m P8 - adversarial board review

Authored 2026-08-08 by the P8 `verify-reviewer` (fresh context, no routing or fixing history inherited). Inputs: `kicad/rf-de-20m.kicad_pcb` as committed, `reports/gate-verify.json` (31 err / 159 warn), `reports/gate-drc_routed.json` (55, 0 unconnected), `reports/route-notes.md`, `reports/spiral-design.md`, `reports/verify-waivers.md`, `architecture/*`, `parts/*`, and renders re-made from the CURRENT board (`reports/renders/rf-de-20m_{top,bottom,iso}.png` - the pre-existing set was 7 h older than the P8 copper edits).

Everything below was measured on the actual board geometry with `lib/geom.py`, or re-derived from the datasheets. Scripts are quoted so the numbers can be reproduced.

Result: 5 errors, 3 warnings. All 8 existing waiver classes are recommended for approval, **six of them unconditionally and two with conditions** (s3). The errors are things no check on this board can see.

—

1. The audit you asked for first: is a real creepage violation hiding?

No. Verified item-by-item, and the waiver's claim is exactly right.

`check_creepage` was re-run standalone and its full per-pair summary read (not just the failing list). Across **all 5 declared HV nets, all 3 `voltage_pairs`, all 4 copper layers and every other net on the board, only THREE (net, net, layer) combinations produce a single item pair closer than the largest applicable IPC-2221 requirement**:

primary	other	layer	min gap	raw pairs under req
‘/SW‘	‘/stage/GATE_Q1‘	F.Cu	0.350 mm	3
‘/SW‘	‘/stage/GATE_Q2‘	F.Cu	0.350 mm	4
‘/SW‘	‘GND‘	F.Cu	0.350 mm	40

Every other pair - including `/tank/TANK_A` at 180 V against all 16 other nets, `/tank/RFOUT` at 145 V, `+40V` at 51 V, and the three `voltage_pairs` (`/SW-TANK_A` at **203 V**, `TANK_A-TANK_B` 151 V, `TANK_B-RFOUT` 135 V)

- reports `min_gap_mm: null`, i.e. **no copper anywhere on the board is even

within the requirement distance**. The 47 raw hits dedupe (0.1 mm spatial) to the 21 errors + 3 warnings reported.

Then, per finding, the two colliding items were dumped by TYPE:

- **21 of 21 involve only EPC2019 pads and the 0.25/0.45 mm die-escape tracks.** Not one involves a **zone fill**. The composition is 8 pad-to-track, 11 track-to-track (drain column vs source/gate escape) and 2 pad-to-track at 0.389/0.391 mm.
- **All 21 gap midpoints lie strictly inside the two die pad bounding boxes** (Q201 `[36.810, 61.885, 39.460, 62.585]`, Q202 ‘[36.810, 66.085, 39.460, 66.785])`, split **Q201 x10 / Q202 x11** - the split` verify-waivers.md‘ claims.
- The 0.350 mm figure is the land pattern's own arithmetic: 0.600 mm column pitch minus the 0.250 mm bar width. 0.375/0.389/0.391 are the round-bump diagonals.

And the pour is clean, measured directly. The `/SW` zone fill's minimum distance to *any* GND copper on F.Cu is **0.8004 mm** (to pads 0.8005, to tracks 0.8005, to GND zone fill 0.8004) - the `aiee_hv_143v_SW` rule honoured to 0.4 um. There is no board-copper creepage violation hiding in the set.

Verdict: waiver entry 1 (21 `check_creepage`) and the matching `drc_routed` clearance class (48) are sound. Approve. The caveat `verify-waivers.md` already carries - 10 of the 21 have empty `refs`, so the subset matcher cannot exclude a future empty-refs finding elsewhere - is the correct compensating control and should be kept: **if the count ever exceeds 21, read the delta, do not waive it.**

—

2. The five errors

E1 - the thermal budget rests on theta_BS = 1.5 C/W for the pair, and that number is optimistic by 2.5-4x. At my estimate the EPC2019 exceeds its 150 C absolute maximum. (error) This is the worst thing on the board and it is the one thing `verify-waivers.md` entry 6 asserts rather than measures.

The waiver's premise is correct. I read `check_thermal.theta_ja(): tfloor + (t0 - tfloor)*exp(-A/tau)` with `A_SAT_MM2 = 645.0`. Copper area is its only input; there is no heatsink, TIM or airflow term. The "645 mm2" in the message is the model's **saturation cap**, not a measurement of this board's island. So yes - the check is heatsink-blind and its 288 C is meaningless here.

The substitute numbers are what fail. The claimed path is `theta_JB 3.75 + theta_BS 1.5 + theta_HS 0.7 = 5.95 C/W` for the pair. `theta_JB` is right (EPC2019 `RthJB_typ` 7.5 C/W per die, halved for two in parallel - `parts/C2836675.json`). `theta_HS <= 0.7` is a hard, measured, external requirement (HS-1) and is out of my scope. **`theta_BS` - F.Cu under the dies through the board to B.Cu - is assumed, and the board's own stackup says it cannot be 1.5.**

Stackup read off the board: F.Cu 35 um / **0.2444 mm** / In1 **15.2 um** / **1.065 mm** / In2 15.2 um / 0.2444 mm / B.Cu 35 um. The 1.065 mm core between two **half-ounce** planes is the dominant term and it is crossed by only **18 vias** (9 per FET, measured - see E5).

Two bounding estimates, FR4 through-plane k = 0.3-0.4 W/mK, copper 400 W/mK:

```
| model | R(F.Cu->B.Cu), pair |
|---|---|
| **optimistic** - heat magically spread over the full 645 mm2 island at every
layer, k = 0.4, in parallel with 18 barrel vias (12.4 C/W incl.  lateral
access) | **4.05 C/W** |
| **realistic** - exponential spreading length per layer (‘lambda =
sqrt(k_cu.t_cu.h/k_fr4)‘ = 3.38 / 4.65 / 2.23 mm), effective areas 110 / 392 /
589 mm2, k = 0.3 | **7.3 C/W** |
```

Even the optimistic bound is **2.7x** the assumed 1.5 C/W. Feeding it back:

```
| theta_BS | theta_JA(pair) | Tj at 11.25 W nom | Tj at 14.6 W max-corner |
|---|---|---|---|
| 1.5 (assumed) | 5.95 | ~123 C (as claimed) | ~142 C |
| **4.0 (optimistic)** | 8.45 | **~151 C** | **~180 C** |
| **7.3 (realistic)** | 11.75 | **~188 C** | **~228 C** |
```

(Ambient 40 C, and the +9 C the waiver's own 6R8 gate-loss adder carries over the `constraints.json` baseline. EPC2019 `TJ_range` absolute maximum is **150 C**.)

`verify-waivers.md` s3 already names this as **OPEN-10** - "theta_BS = 1.5 C/W for the pair is assumed, not simulated, and carries 17 C of the 85 C junction budget". That is the right instinct but the wrong magnitude: it carries far more than 17 C, and it is the term that decides whether this board survives its own design point. **The `check_thermal` waiver must not be approved as "waivable, real numbers are fine" - it should be approved as "the gate is blind, AND the real number is unresolved and blocking."**

Levers, cheapest first, all of which move this materially: 1. **Order 1 oz inner layers** instead of 0.5 oz - halves the In1/In2 lateral spreading resistance, an order-time option, no layout change. 2. **Build the missing B.Cu mask-opened land** (E3) and get the vias fixed (E5). 3. **Derate the bus to 36 V / 162 W** - already costed by the architecture at **-14 C**. 4. **Revert R205/R206 to 4R7 - -4 C**, and W1 below shows the threat that bought 6R8 was over-estimated ~5x. 5. Close OPEN-10 with a 3-D thermal solve or a first-article measurement **before ordering**, not after.

E2 - the C_m bank ships 8.8 % high because the mandated RFOUT pour capacitance was never folded in. The as-built match is WORSE than the 560 pF bank P4 explicitly

rejected. (error) P7 measured the pours' added capacitance to GND (`route-notes.md` s7) and handed the populate decision to P8. `verify-waivers.md` s16 records the outcome: *"the s7 populate recommendation is untouched."* It was never taken.

I re-derived the number independently rather than trusting it: the `/tank/RFOUT` F.Cu pour is **371.5 mm2**, of which **84.5 % (313.9 mm2)** sits over In1 GND at 0.2444 mm. Parallel plate at the board's declared `epsilon_r = 4.05` gives **46.1 pF** (48.9 pF at 4.3). P7's figure was 46.06 pF. Confirmed.

That 46.1 pF is a lumped shunt at the RFOUT node at 20 MHz - **directly in parallel with C_m and the 50 ohm load**. So:

```
| C_m | Z presented to the tank | series X/R | P_out at 40 V |
|---|---|---|---|
| 484 pF physical + 46 stray = **530** (the ideal) | 4.13 + j0.00 | 1.153 | 200 W
|
| **531 pF populated, stray ignored** (the BOM's own solve) | 4.12 + j0.07 |
1.19 | ~200 W |
| 560 pF - **rejected at P4 review W1** | 3.74 + j0.67 | 1.47 | ~221 W |
| **531 + 46.1 = 577 pF, AS SHIPPED** | **3.53 + j1.01** | **1.65** | **~233 W**
|
```

(`omega L_s` 20.607 - `1/omega C_s` 15.790 + X_load, R_load from `50 || 1/jwC_m + jwL_m`; reproduces both published anchors exactly.)

P4 review W1 added a whole new BOM line (`C541492`, 27 pF) for the sole purpose of getting the bank off 560 pF*, *on the argument that 221 W* "inflated I_dc, FET conduction loss and the 11.25 W / Tj 114 C two-FET thermal budget the whole paralleling decision rests on"* (`parts/parts.json`, C541492 role note). The board as it stands lands **past that rejected point: ~233 W, +17 % I_dc, **+37 % FET conduction loss**, on the board of E1 above.

Fix is one BOM line and it is already specified: depopulate one 56 pF site from C_m (C310-C318 -> 8 populated), keep C319. 8x56 + 27 = 475 pF + 46.1 = **521 pF, -1.8 % on the ideal** - exactly P7's recommendation. Domain `parts`.

Two smaller siblings, folded in for completeness and NOT separately reported: the `/tank/TANK_A` pour adds 13.4 pF from the mid-node to GND (shunts ~2.7 % of the tank current - reactive, so it detunes rather than dissipates), and the two spirals' mutual at the as-placed 46.8 mm centres is about **-1.1 nH each, -0.8 % on the 274 nH series tank** (`spiral-design.md` s6, which asks P8 to "fold the number in rather than ignore it" - also not done).

E3 - there is no mask-opened heatsink land on B.Cu. There is no solder-mask aperture on B.Cu anywhere. (error) `architecture/stackup.md` ("Thermal vias under the FETs") and `constraints.json:thermal[Q201]` both require *"mask-opened land on B.Cu"*.

Measured on the board: the **only** layer sets that reach a mask layer are `"F.Cu" "F.Mask" "F.Paste"` (161 component pads) and `"*.Cu" "*.Mask"` (6 items = J101's two THT pins + the four NPTH mounting holes). **Zero B.Cu-only mask apertures exist.** `reports/renders/rf-de-20m_bottom.png` confirms it visually - the whole bottom face is green, including the entire HS-2 rect `[5, 10, 36, 70]` board-local.

So the sink presently bolts onto **solder mask**, which nobody decided. This is a fork the owner has to rule, not something to silently accept:

- **Keep the mask** - then say so, and add the mask term (~25 um at k ~0.25 W/mK; 0.05 C/W if the contact really is 1860 mm2, **0.5 C/W** if the effective spreading area is only ~200 mm2) to the E1 budget explicitly.
- **Open it** - then the **HASL -> ENIG override already in `stackup.md` becomes load-bearing, not a nicety**: HASL on a 31 x 60 mm bare-copper land leaves 5-25 um of solder relief and destroys the flatness the whole thermal case assumes.

Either way, this is an unimplemented, explicitly-stated architecture requirement on the path that carries most of the junction budget.

E4 - nothing on this board can clamp the heatsink over the FETs. (error) HS-1 requires **theta_HS <= 0.7 C/W measured, in forced air, over a ~31 x 60 mm base**, and E1 shows the interface resistance is where this design's margin lives. The board provides no way to make that interface.

- Mounting holes are `4x M3 (3.2 mm) at corners`, board-local (3,3), (117,3), (117,77), (3,77) - the P0 Q4 default, chosen before the heatsink existed.
- **H2 and H3 are at x = 117.** HS-2 forbids the sink past x = 40 mm, so they cannot hold it.
- That leaves **H1 and H4 only, both at x = 3, collinear on the west edge** - two bolts on one line, 28-30 mm away in x from the dies at (31.5, 22.9) and (31.5, 27.1). A base clamped on one edge cannot control contact pressure at the far corner of a 31 x 60 mm face, and pressure is what sets the TIM bondline.
- There are **no holes inside the HS-2 rect** and nothing in `blocks.md`, `stackup.md` or `requirements.md` specifies a fastening scheme at all - "fastener" appears once, in SPIRAL-6, only as a thing to keep away.

I checked that adding holes is geometrically legal: with the sink confined to `x <= 40`, SPIRAL-6 (no metal within 15 mm of a spiral's outer copper edge) is satisfied with 19.4 mm to spare, and the existing corner holes are 18.8 mm and 30.8 mm clear of L302/L301. Recommendation: **2-4 M3 NPTH holes inside or on the boundary of `[5, 10, 36, 70]`**, placed to miss the power loop and the In1/In2 planes' critical area, or an explicit clamp-bar/spring-clip scheme recorded as a mechanical deliverable. Owner decision at H4.

E5 - 9 thermal vias per FET against a required >= 10, none nearer than 1.1 mm to a source land, and "copper-filled" is not a JLC process. (error) Measured from each die centroid:

	vias <= 2.27 mm	vias <= 4.0 mm	nearest
Q201	3	**9**	1.581 mm
Q202	3	**9**	1.598 mm

The `check_thermal thermal_vias` warning's triage in `verify-waivers.md` s3 is **arithmetically correct** - the check's window really is 2.27 mm and the real count really is 9 - but it then argues 9-vs-10 is worth "<0.1 C". Under E1's numbers the via array is not a "parallel helper worth a few percent": with the 1.065 mm core in the bulk path, **18 barrel vias at ~187 C/W each carry roughly half the total heat flow.** Two things follow:

1. `blocks.md` s4.1 and `stackup.md` require **">= 10 x 0.3 mm COPPER-FILLED vias per FET, in and immediately around the source lands"**. The board has 9, and the nearest is 1.58 mm from the centroid (~1.1 mm from the nearest source pad) because `aiee_hv_143v_SW` at 0.8 mm caps the landing area. Neither the count nor the "immediately around" is met. 2. **JLC does not sell copper-filled vias.** The order-time option in `stackup.md` is **POFV - epoxy filled and capped** (specified there for the LMG1020 balls). Resin fill is not copper: the "filled, not plated-barrel-only (~2.5x worse)" factor the thermal budget assumes **is not purchasable on this vendor**, and no one has said so. Carry to P10 as a hard constraint on the thermal case, not as a nice-to-have.

Note also that the LMG1020 via-in-pad that POFV was requested for **was never built** - the nearest GND via to U201 is at (30.777, 63.315), 0.3 mm outside the ball array, not in a pad. That is consistent with, and part of the cause of, the `check_decoupling` loop warnings (s3).

—

3. Warnings

W1 - common-source inductance was escalated from P7 to P8 and was never closed; my estimate says it is survivable, and that the 4R7 -> 6R8 change was bought against a ~5x over-estimate (warning) `route-notes.md` s14 raises it explicitly - *"This deserves a P8 SIM item - it was never modelled by the architecture"* - and `verify-waivers.md` does not mention it anywhere. Closing it here.

The source escapes were measured (all 0.25 mm wide, F.Cu, over In1 at 0.2444 mm, `mu0.h/w` = 1.23 nH/mm):

pad	escape	L
2 (gate-return star point)	0.400 mm west	0.49 nH
6	0.450 mm east	0.55 nH
7	0.450 mm east	0.55 nH
4	1.200 mm outward	1.47 nH

The die's internal source metal ties all four, and **all four land on the same F.Cu GND island** (one 311.7 mm2 island that also carries U201's ground and C203/C204's ground - verified). So the inductance common to the gate loop and the power loop is the **parallel combination, ~0.157 nH**, not the 0.768 nH single escape P7 used. That changes the answer by 5x:

- turn-off, per FET: `I_off ~5 A / t_f ~2.1 ns -> di/dt 2.4 A/ns` -> **+0.38 V** on VGS (worst case at P7's 8 A/ns pair figure: **+0.63 V**);
- Miller: `Crss 0.7-1.0 pF x dV/dt 24.8 V/ns` = 17-25 mA into ~8 ohm -> **+0.14-0.20 V**;
- total **+0.5 to +0.8 V** against `VGSth_min` **0.8 V** (typ 1.4 V).

Not a destruct path, but only ~1.3x margin at the datasheet minimum, and it is the mechanism a second-order RLC model cannot contain. It should be a named bring-up measurement (VGS at the die during the turn-off edge), not an assumption. There is also a layout answer if it ever bites: pad 2's escape is currently merged into the shared GND pour 0.4 mm from the die, i.e. **there is no Kelvin source** - routing pad 2 separately to U201's ground would remove the term entirely, at the cost of a re-place.

Consequence for R205/R206. P7 chose 6R8 over 4R7 partly on *"a `L_common.di/dt` term of the order of volts"*. At 0.157 nH it is 0.4-0.6 V, not volts. On the published numbers 4R7 already passes both rails (-0.85 V against -4 V; +5.005 V against +6 V). **Reverting to 4R7 recovers ~4 C of the E1 budget for no new risk** - worth taking if E1 stays tight. This is a recommendation, not a defect; the 6R8 decision was ruled twice and is the owner's.

W2 - C_shunt ships at 455 pF, 1.4 % above the top of the design's own 403-449 pF band (warning) Same root cause as E2 and the same abandoned P7 handoff. `stage.py`'s recorded deviation solves `316 (pair Coss) + 112 (2 x 56 pF) = 428 pF`, "the MIDDLE of the 403-449 pF requirement". It predates P7's measurement of **+27.4 pF** of `/SW` pour capacitance to In1. As shipped: **455.4 pF**, above the band, and 13 % above the ideal-choke value.

Much less serious than E2, because the trim is real and downward: depopulating C204 gives 316 + 56 + 27.4 = **399 pF**. **Ship it as a bring-up instruction** ("if ZVS lands late, pull C204 first"), or take P7's ~60 pF recommendation now. Domain `parts`.

W3 - no fiducials anywhere on the board (warning) `grep -i fiducial` over the .kicad_pcb returns nothing; the renders confirm it. This board carries a **0.4 mm-pitch WCSP** (U201, 1.0 x 0.6 mm ball array) and **two bare passivated die** whose lands are 0.23 mm solder-mask-defined openings with 0.25 mm bumps - the three finest-pitch things a JLC PCBA line will see on this order. JLC will use panel fiducials, but local fiducials are the standard mitigation below 0.5 mm

pitch and cost nothing. Recommend **3 x 1 mm copper / 2 mm mask fiducials** on F.Cu, in the empty area around board-local (10, 10) / (110, 10) / (10, 70), all outside the HS-2 rect and >= 15 mm from either spiral. Domain `fab`.

—

4. Verdicts on the rest of the machine output

Every residual class, with a verdict. Nothing here needs re-work.

class	count	verdict
‘check_creepage‘ 21 err + 3 warn	24	**Waiver sound - independently verified, s1.** Approve, keep the "count is fixed at 21" control.
‘drc_routed‘ clearance, Q20x	48	Same geometry, same evidence. **Approve.**
‘check_current‘ ‘undersized_track‘ ‘/SW‘	3	**Waiver sound, and the consequence is benign - I measured it.** The 0.25 mm drain escapes are only ~1.0 mm long: the ‘/SW‘ **pour reaches x = 37.42 into the inter-die gap and is 9.58 mm wide continuous from y 63.4 to 65.0**, so the constricted section is 2 x 0.25 mm x ~1.0 mm per FET, 1.31 mOhm, ~55 mW, and a both-ends-clamped fin gives **dT ~2 C** (‘P.L/(8.k.A)‘). IPC-2152 does not apply to a 1 mm neck. Approve.
‘check_current‘ ‘undersized_track‘ ‘/tank/RFOUT‘	1	Geometrically unmeetable (9.106 mm corridor, 0.8 mm HV each side). **Approve.**
‘check_current‘ ‘pour_neckdown‘ ‘+40V‘	1	Per-zone erosion artefact at 3 of 30 vias; the P8 resistive-sheet solve is the better evidence. **Approve.**
‘check_return_path‘ ‘/SW‘	1	**Waiver sound - re-measured independently.** ‘/SW‘ pour 191.30 mm2, of which **183.90 mm2 = 96.13 % over continuous In1 GND** (matches the claim to 2 dp), and the whole loop sits on **one** In1 island (In1 has 3 islands - zone A 3931 mm2, zone C 1637 mm2, a 0.3 mm2 sliver - and the drain node, C_shunt, U201 and the gate legs are all on zone A). GATE_Q1/GATE_Q2/GATE_ON/GATE_OFF are **100.0 %** imaged; DRIVE 96.9 %. The deficit is the L301 land at the zone-B boundary, exactly as argued. **Approve.**
‘check_thermal‘ Q201/Q202	2	**Approve the waiver on its premise (the check is heatsink-blind - confirmed in source), but NOT on its numbers.** See E1.
‘check_thermal‘ L301/L302	2	**Approve.** The winding is a pad, so the check measures the tank pour, not the conductor; ‘spiral-design.md‘ s4 is authoritative and its 2.99 / 2.48 mW.mm-2 against SPIRAL-1’s 7 is 2.3x clear. Independently sane: ‘P = 200.Q_L/Q_ind‘ reproduces 2.58 / 2.05 W.
‘check_current‘ 119 ‘insufficient_transition_vias‘ + 13 ‘pour_neckdown‘ + 8 ‘undersized_track‘, all derived-GND	140	Synthesised return-net entry at the largest rail’s 7.0 A. Documented labelled worst case. **Waive.**
‘check_decoupling‘ 4 distance + 2 loop (C201 6.1 nH, C202 3.1 nH vs 0.3 nH)	6	**Real, correctly attributed to the P6 backward edge, and low-consequence HERE - which the waiver does not say and should.** I traced C202’s link: 2.49 mm of 0.2 mm track, no vias, ~3.8 nH by ‘mu0.h/w‘, so the estimator is right. But the VDD loop is only in the **turn-ON** path (turn-off sinks through OUTL to GND), and Class E turn-on is ZVS - the drain is already at 0 V, so a ~2.3 ns edge (zeta 0.85 at L 5.55 nH, R 9 ohm, C_GS 199 pF) costs almost nothing. **Waive, with that reason recorded.**
‘check_creepage‘ 3 warnings	3	Pad-to-pad inside one EPC2019 land pattern; the check labels them itself. **Waive.**
‘check_thermal‘ ‘thermal_vias‘	2	Count triage correct; conclusion wrong. **See E5.**
‘check_silk‘ ‘silk_misattributed‘	6	Real, unfixable for R203 (0 legal positions found by the P8 grid search), and JLC places from the CPL. **Waive.**
‘check_pdn‘ ‘pdn_no_bulk‘ ‘+40V‘	1	Sidecar scope artefact; C101/C102 are the bulk. **Waive.**

| ‘verify_all‘ ‘constraints_drift‘ | 1 | Records question. **Waive, and take the P8 recommendation to retire ‘architecture/constraints.json‘** - a second copy in a different coordinate frame is how the P6 bug happened. |
| ‘padstack‘ x2, ‘copper_sliver‘ x1 | 3 | Deliberate inner-only bridge pad; unlocatable sliver. **Waive.** |

No check was SKIPPED. All 8 `verify_all` members ran (`by_check` in `gate-verify.json` lists every one, `check_diffpair` legitimately 0).

But a different hole exists, and no gate can see it: none of SIM-1..SIM-6 has ever been run. There is no `kicad/sims/`, no sim report, nothing in `state.json` (`open_issues: []`, `gates: {}`, phase still `P7`). `decisions.md` declares **SIM-4 ‘tank_match‘ at ERROR severity** and calls it *"the C_s arbiter"* (OPEN-12); SIM-1/SIM-2 are the arbiters of the 403-449 pF C_shunt band. `verify_all` has no `sim_run` member, so a board can reach a green verify gate with every declared simulation gate unexecuted. **E2 and W2 are exactly the questions SIM-4 and SIM-2 exist to answer.** Recommend the owner treat "SIM-4 unrun" as blocking alongside E1/E2 at checkpoint 4.

—

5. Things I checked that are FINE (so nobody re-checks them)

- **Tank capacitor family and rating.** `CC1206JKNP0CBN560` / `CC1206JKNP0CBN270`
  - **1 kV C0G (NP0), 1206, no X7R anywhere in the tank** (X7R appears only on the buck rails and the driver VDD, where it belongs). 1 kV against the 151 V pk across the C_s bank is 6.6x; against TANK_A's 180 V node, 5.5x.
- **Tank current sharing.** The banks are parallel arrays of identical parts, so sharing follows capacitance: 0.77 A rms per 56 pF part in C_s, 0.70 A in C_m, 0.34 A in the 27 pF. The C_s column is 24 mm long, but its series inductance (˜0.7 nH, j0.09 ohm) is 0.06 % of each cap's 142 ohm - **sharing is not degraded by the column geometry.** Bank ESR loss is ˜1.0-1.1 W each at C0G Q ˜700, which is real and is not itemised in `power_tree.md`; it is inside the 27-39 W envelope, so noted not reported.
- **Gate-leg symmetry.** GATE_Q1 and GATE_Q2 are **identical to the micron**: three segments, 1.000 + 0.989 + 1.451 = 3.440 mm each, widths 0.400 / 1.000 / 1.000, mirrored about y = 64.110 - which is exactly the midpoint of the two **gate pads** (62.010, 66.210), so the mirror lands correctly despite the two dies being translated rather than mirrored. Both legs are **100 % over In1 GND**. The die-local return geometry is identical (pad 2 at +0.450 mm from pad 1 on both, 0.400 mm west escape on both). The +/-0.1 nH matching spec is met by construction.
- **The drain escape** - see s4; measured, ˜2 C, benign.
- **The ‘/SW‘ power loop.** One 263 mm2 F.Cu island, 96.13 % imaged on In1, the C_shunt bank and L202.1 100 % imaged, In1 unbroken beneath the whole loop. The C203/C204-populated / C205/C206-DNP asymmetry costs Q202 about **4 mm of extra travel in a 9.58 mm-wide pour = ˜0.13 nH** - immaterial, because the stacked-loop architecture is doing its job.
- **Spiral encoding.** All spiral pads are copper-only (`"F.Cu"`, `"B.Cu"`, `"In1.Cu" "In2.Cu"`, and 84 `"*.Cu"` plated holes) - **no F.Mask and no F.Paste anywhere on either spiral.** The windings are fully mask-covered and no stencil aperture will ever be cut over them. This was the failure mode I went looking for; it is not present.
- **B.Cu spiral vs the sink.** SPIRAL-6 holds: the heatsink land ends at x = 36 board-local and L301's outer copper starts at 55.45 - 19.4 mm clear. The corner mounting holes are 18.8 mm (H3/L302) and 30.8 mm (H2/L301) clear.

- **HS-3.** J101 is the only THT part, at board-local x = 3.7, outside the `[5, 10, 36, 70]` rect. Its two `"*.Cu" "*.Mask"` pads are the only B.Cu copper the sink could touch, and they are clear.
- **Via tenting.** No pad or via on the board declares a B.Mask aperture, so the six non-GND vias inside the heatsink land (2 x `+5V`, 2 x `+5V_DRV`, 2 x `/stage/DRIVE`) are tented by construction. The P7/P8 FAB NOTE ("do not enable via-tenting-off at plot time") remains the right control - carry it to P9.
- **EPC2019 land pattern.** Copper = bump size, mask -0.01 mm/side giving the datasheet's 0.23 / 0.23 x 0.68 mm openings, paste the same. Pad-4 escapes run **away** from the inter-die gap (north from Q201, south from Q202), which is why the gap is clean. A pin-1 dot exists on F.SilkS at (-1.40, -0.63).
- **LMG1020 input abs-max** with a bench generator - already found and fixed at P4 (review E3). Not re-reported.
- **Minor, prose only:** the EPC2019 courtyard is `2.76 x 0.94 mm` - **exactly the die envelope, zero IPC-7351 excess.** Nothing collides today (nearest neighbour C203 is 1.3 mm away), but any future re-place gets no warning. Also, the round 0.23 mm pads' stencil area ratio is 0.58 at 0.1 mm foil, under the usual 0.66 - acceptable **only** because the EPC2019 arrives with its own solder bars and the paste is largely a flux carrier. Worth knowing at first article, not worth a finding.

—

6. Reproduce

.venv/Scripts/python .claude/skills/ai-ee/scripts/check_creepage.py \ –pcb boards/rf-de-20m/kicad/rf-de-20m.kicad_pcb \ –constraints boards/rf-de-20m/kicad/constraints.json –out creep.json # then read checked[].pairs[].min_gap_mm - only 3 pairs are non-null

.venv/Scripts/python .claude/skills/ai-ee/scripts/render.py \ boards/rf-de-20m/kicad/rf-de-20m.kicad_pcb –views top,bottom,iso \ –w 2400 –out-dir boards/rf-de-20m/reports/renders

Geometry measurements (pad bboxes, zone-fill distances, the inter-die pour probe, In1/In2 island split, the RFOUT-over-In1 area) were made with `.claude/skills/ai-ee/scripts/lib/geom.py` `load_board()` + `net_copper(net, layer)` / `zones_of()` / `pads_of()`; the exact probes are quoted inline above.

8 DFM and Fabrication

Gate `dfm` (phase P9): **pass** after 1 attempt(s); last 2026-08-09T00:14:32, failing 0 of 0. DFM findings by severity: none.

Fabrication outputs

Exported layers: F.Cu, In1.Cu, In2.Cu, B.Cu, F.Silkscreen, B.Silkscreen, F.Mask, B.Mask, F.Paste, B.Paste, Edge.Cuts.

Bill of materials

Comment	Designator	Footprint	LCSC
100uF 63V alu electrolytic	C101,C102	CAP-SMD_BD10.0-L10.3-W10.3-LS11.0-FD_1	C51953411
2.2uF 100V X7R 1206	C103,C104	C1206	C577211

Comment	Designator	Footprint	LCSC
100nF 100V X7R 0603	C105,C109,C111	C0603	C113803
2.2uF 50V X7R 0805	C106	C0805	C125847
10nF 100V X7R 0603	C107,C207,C208,C209,C210	C0603	C107059
22uF 16V X7R 1206	C108	C1206	C22392398
1nF 100V C0G 0603	C110,C211,C212	C0603	C113793
100nF 16V X7R 0402	C201	C0402	C60474
10nF 25V X7R 0201	C202	C0201	C285010
27pF 1kV C0G 1206	C204,C319,C320	C1206	C541492
1uF 16V X7R 0603	C213	C0603	C106248
56pF 1kV C0G 1206	C301,C302,C303,C304,C305,C306,C307,C310,C311,C312,C313,C314,C315,C316,C317	C1206	C121873
120R@100MHz bead 3A 30mohm	FB201	L0805	C79382
KF128-5.08-2P 2pos 5.08mm 24A/250V	J101	CONN-TH_P5.08_KF128-5.08-2P_EDGETRIM	C474952
SMA jack 50R SMD (CONSMA001)	J201,J301	SMA-SMD_CONSMA001-SMD-G-T	C22418168
22uH 1A shielded	L101	IND-SMD_L4.0-W4.0_SLW4010S	C83472
470nH 20A 4.1mohm	L201,L202	IND-SMD_L7.0-W6.6_FXL0630	C167212
EPC2019 200V eGaN	Q201,Q202	TRS-SMD_EPC2019	C2836675
30.9k 0402 1%	R101	R0402	C270615
10.0k 0402 1%	R102	R0402	C25744
100k 0805 1% 150V	R103,R104	R0805	C149504
100R 2010 1% 750mW	R201,R202	R2010	C421954
4R7 0805 1% 250mW	R203,R204,R205,R206	R0805	C160081
LM5017 100V sync COT buck	U101	SOIC-8-EP_LM5017_TI-MRA08B	C34355
LMG1020 5V GaN driver 7A/5A	U201	BGA-6_L1.3-W0.8-P0.40-TL_PTMAG3001A2YBGR	C6423790

1 CPL rotation correction(s) applied; no missing LCSC numbers. *fab/order.json not available.*

9 Run Record (Appendix)

Phase digests

P0 *(no digest recorded)*

P1 *(no digest recorded)*

P2

```
# P2 Architecture digest - rf-de-20m (20 MHz Class E GaN stage, 200 W)

**AMENDED 2026-08-07 (P2-A)** after the orchestrator read the EPC2019 datasheet
(rev (c)2021) directly and retracted two P1 figures.  This digest reflects the
amended package.

Artifacts:  ‘architecture/blocks.md‘, ‘power_tree.md‘, ‘stackup.md‘,
‘sheets.md‘,
‘constraints.json‘, ‘decisions.md‘.

---

## AMENDMENT P2-A - a retraction, plus two further errors found while
re-deriving
```

RETRACTED. ‘research/power.json‘ supplied **Coss(er) 156 pF** and **Rds(on) 22 typ / 42 max mohm**. The datasheet publishes **no Coss(er), no Coss(tr) and no Eoss anywhere** - the 156 pF was invented - and Rds(on) is **36 typ / 50 max mohm**.
The original ruling *"2 x Coss(er) = 312 pF lands on the required 317 pF, so the external shunt cap disappears"* is void.

ENDORSED. The orchestrator’s method is right: **‘Coss(tr) = Qoss/V‘ (charge-equivalent) is the correct basis for Class E**, because the drain waveform is produced by integrating current into charge. ‘Coss(er)‘ is the *hard-switching energy* equivalent (~119 pF here, 25% low) and the 110 pF small-signal figure is quoted at 100 V where Coss has already fallen. Both would have under-sized the shunt.

ERROR 1 - wrong voltage range. ‘18 nC/100 V = 180 pF‘ is the average over 0-100 V; this drain swings to **142.5 V** and Coss keeps falling. Two independent extrapolations (power-law fit anchored on *both* published numbers, and a flat-above-100 V bound) **agree to 1.5%: Coss(tr) = 158 pF typ / 205 pF max**.

ERROR 2, the bigger one - wrong Sokal coefficient. ‘0.1836/(omega.R)‘ is the **Q_L -> infinity** constant; this board runs **Q_L = 5**, and ‘refdesign-classE-stage.json‘ D9 already said so. Sokal’s published fits reproduce the fragment’s power and shunt coefficients **to 0.008% and 0.13%**, confirming both. **The frozen operating point itself was computed with Q_L -> infinity constants on a Q_L = 5 design.**

	Frozen	Orchestrator’s fix	**P2-A**
Vdd	40 V	37.5 V	**40 V - frozen input preserved**
R_opt	4.614 ohm	4.06 ohm	**4.13 ohm**
C_shunt required	317 pF	360 pF	**403 pF** (+38-46 pF finite-choke term)
Coss(tr) per FET	(invented 156)	180 pF	**158 typ / 205 max**
Pair supplies	312 pF	360 pF	**316 typ / 410 max**
External C0G trim	5 pF	0 pF	**87 pF, 0-133 pF of range**

The two errors pull opposite ways and the second dominates: the bus stays at 40 V, and the external trim capacitor comes back. That last point is the most valuable outcome - the original ruling treated its disappearance as elegance, but it is a **defect**: Coss spread on this part is 110-150 pF (+36%), and the external cap is the **only** absorber. At 100% device shunt a max-Coss part is unfixable without reworking etched copper.

The four rulings, as amended

1. **TWO EPC2019 - REINFORCED, and now absolute.** With the real Rds(on), a single FET reaches **Tj 160 C with a HYPOTHETICAL 0 C/W heatsink - above the 150 C

absolute maximum.** There is no heatsink, TIM, via array or copper weight that saves a one-FET build (it was "138 C against a 125 C target" before). The pair:
114 C nominal / 133 C at the max-datasheet corner at theta_HS 0.7 C/W.
The "k = 2 is exactly the ZVS ceiling" sub-argument is RETRACTED (the real ceiling is 2.55) and replaced by a quantitative one:
‘P_FET(N) = 5.48/N + 2.42N + ...‘ gives **11.17 / 11.25 / 13.16 W at N = 1/2/3**.
Paralleling does not reduce dissipation at all - it halves thermal resistance.
N=3 pays +1.9 W and a three-way gate match that cannot be mirrored.
2. **C_shunt = 403 pF: 316 pF from the pair + ~87 pF external.** C203-C206 are four
1206 C0G sites, **3 populated (99 pF), no longer DNP**, in the power loop.
3. **PCB spirals, re-derived: L_s 164 nH, L_m 110 nH.** The **losses barely moved**
(1000/Q and 666/Q) because ‘P = 200.Q/Q_ind‘ is **independent of R** - the tank current rises exactly as fast as the reactance falls. The lower L is geometrically helpful: 164 nH at the same OD wants a *wider* trace, which is what
Q and the thermal area want anyway.
4. **Two-zone floorplan - unchanged.** The power loop *ends* where the tank *begins*, so the two plane rules apply to disjoint regions.

New findings at P2-A

- **The frozen tank was wrong for its own Q_L.** R 4.614 -> **4.13 ohm**; C_shunt 317 -> **403 pF**; L_s 184 -> **164 nH**; C_s 447 -> **518 pF +/-5%**; L_m 115 -> **110 nH**; C_m 500 -> **530 pF**; I_tank 6.58 -> **6.96 A rms**. Tank conductor widths grow ~12% (7.2 mm, was 6.4).
- **A third P1 coefficient is refuted and NOT adopted.** The fragment’s ‘C_series.omega.R = 0.63467‘ implies a net series reactance of **3.4 R** - not a
Class E network. Sokal’s fit gives **0.26906 -> 518 pF and 1.283 R**, which also
agrees within 3.5% with the independent ‘X_net = 1.1525 R‘ relation.
SIM-4 gained a measure (‘x_net_series_over_r‘) specifically to arbitrate it.
- **Vdd is NOT a ZVS knob - but it IS a free thermal derating knob.** ZVS in Class E
is a property of the *network*: the equations are linear in Vdd and the ZVS/ZdVS
conditions constrain only the current waveform shape. So the bus can be backed down at bring-up without losing ZVS: **40 V/200 W/Tj 133 C -> 36 V/162 W/Tj 119 C**
at the max corner, zero rework. This is the mitigation for the compounded worst
case (170 C), and it corrects the orchestrator’s framing - lowering the bus alone
cannot fix a shunt-capacitance error.
- **The gate-loop budget TIGHTENED to 0.48 nH per FET** (was 0.84). The datasheet’s
Ciss 200 pF / Crss 0.7 pF give C_GS ~ 199 pF, not the ~350 pF the retracted Qg implied. **Now the tightest layout spec on the board**, with a stated fallback (R_G 3 ohm -> 0.84 nH, at +0.3 W turn-off loss).
- **The real Qg of 1.8 nC IMPROVED the +5 V rail**: 143 -> **99 mA**, 3.0x headroom,
and gate-resistor dissipation 0.047 -> 0.029 W per part.
- **HS-1 tightened: theta_HS <= 0.7 C/W** (was 1.4). A passive bolt-on will not do it.
- **BOM +$4/board**: EPC2019 is **OUT OF STOCK at LCSC (stock 0), repriced

```
$2.17 -> $3.93**.  Owner approved continuing and **holding the order**; P10
re-verifies.  Delivered estimate **~$32-40/board (~$160-200 for 5)**.

## Unchanged

Stackup ‘JLC04161H-1080B‘ (chosen for its 0.2444 mm L1-L2 gap), 1 oz outer,
TG155 + ENIG + POFV, 100 x 80 mm with pre-authorised growth to 120 x 80, the
two-zone floorplan and plane regions, the relaxed controlled-Z output (D7),
DC-coupled unipolar drive (D8), no protection parts, three sheets and the
net-naming contract.  **Vds,pk stays 142.5 V** - it depends only on Vdd.

## Pipeline traps - two carried, one retired

1.  **‘rules_gen‘ only solves impedance for DIFFERENTIAL pairs.** A
single-ended
‘high_speed‘ entry with ‘impedance_ohm:  50‘ emits **no width rule at all**.
‘/tank/RFOUT‘ is declared under ‘power‘ instead - which is also the right
physics
(**OPEN-2**).
2.  **‘detect_diff_pairs‘ auto-pairs ‘high_speed‘ nets ending ‘_H‘/‘_L‘**, hence
‘/stage/GATE_ON‘ / ‘_OFF‘.
3.  **‘planes_gen‘ has no void or keepout support**; zone B is unpoured by
construction and the spiral courtyards need hand-added rule areas at P6.
**‘board_init‘ does not origin the outline at (0,0)** - translate every rect by
‘outline_bbox‘ at P5.
4.  **RETIRED:** "‘rules_gen‘ puts every power net in ONE class at the widest
width"
is **fixed** in the current script (one class per required width).

## Riskiest decision

**Coss hysteresis is now the dominant FET loss term (4.83 W of 11.25 W) and
rests
on an assumed 10% of stored energy that EPC does not publish** - the same class
of
gap that produced the retracted Coss(er).  At 15% the pair reaches ~13.7 W and
Tj
~127 C. Mitigations are in the architecture at zero cost:  the trim bank, the
bus-derating ladder, and a first bring-up step that settles it in an afternoon
(measure DC input power at 200 W out, thermal-image both dies).

## ‘constraints.json‘

**Lint-clean:  0 errors, 0 warnings.** 4 high_speed, 6 power, explicit empty
diff_pairs, 5 voltages, 3 voltage_pairs, 7 thermal, 6 planes, placement
edges/groups/keepouts/separation/fixed.  Every current, voltage and power
figure
re-derived at R = 4.13 ohm.

## Sim gate - SEVEN benches (SIM-7 added at P2-A)

**SIM-1** ‘classe_zvs_nominal‘ (Vds@turn-on <= 5 V, dVds/dt +/-0.2 V/ns, Pout
190-210 W, Vds pk <= 155 V). **SIM-2** ‘cshunt_sweep‘ - **now validates the
whole
amended basis**, with ‘coss_tr_per_fet_pf‘ bounded to **140-180 pF** from the
vendor
nonlinear model plus a 300-430 pF sweep and ‘trim_pf_needed <= 133‘; must pass
before P4 fixes any tank value.  **SIM-3** ‘gate_symmetry‘ (Id imbalance <= 15%
at
+/-0.15 nH, Vgs inside -4.0/+5.75 V). **SIM-4** ‘tank_match‘ (Zin 3.72-4.54
```

```
ohm,
H2 <= -20 dBc, H3 <= -33 dBc, **plus ‘x_net_series_over_r‘ in 1.0-1.6 to
arbitrate
C_s**, and the spiral-to-spiral mutual k).  **SIM-5** ‘loss_split‘ (pair <=
13.0 W
typ / <= 16.0 W max-corner - the numbers the 0.7 C/W heatsink was solved
against).
**SIM-6** ‘gate_loop‘ (advisory).  **SIM-7** ‘bus_derating‘ **(new)** - re-run
SIM-1
at 40/38/36/34 V with the network unchanged, to confirm ZVS is Vdd-independent
and
the derating ladder is real.
```

P3 *(no digest recorded)*

P4 *(no digest recorded)*

P5 *(no digest recorded)*

P6 *(no digest recorded)*

P7 *(no digest recorded)*

P8 *(no digest recorded)*

P9 *(no digest recorded)*

Gate attempt history

Timestamp	Gate	Status	Failing/Total
2026-08-08T23:51:20	drc_routed	fail	55/55
2026-08-08T23:51:20	sim	fail	1/4
2026-08-08T23:51:20	verify	pass	0/189
2026-08-08T23:51:21	erc	pass	0/0
2026-08-09T00:14:32	dfm	pass	0/0

Issues

(no entries)

Human checkpoints

Id	Phase	Status	When	Note
1	P2	approved	2026-08-07T22:07:56	Architecture approved incl. override of frozen single-FET spec to 2x EPC2019 parallel; unipolar 0/+5V drive; controlled-Z relaxed; outline 100x80 with pre-auth growth to 120x80
2	P4	approved	2026-08-08T10:21:46	Schematic approved (erc 0/0, netlist_audit exit 0, 10/10 review findings fixed). Continue P5-P9 to fab-ready; P10 ordering held pending EPC2019 restock

Id	Phase	Status	When	Note
3	P6	approved	2026-08-08T13:23:42	Placement approved. SMA vertical top-mount (not edge-launch) accepted by owner - it is the only true-SMD 50R SMA at LCSC with no bottom-side joints, which the flat heatsink face requires
4	P8	approved	2026-08-08T21:18:05	Owner approved all 8 waiver classes; owner chose: fix the four board-level thermal defects AND derate to 36V/~162W. Re-confirms schematic approval after the 3 post-H2 human_hold edits
5	P10	not reached		

Generation cost

Not recorded (no reports/cost.json).

10 Artifact Index

File	Size	Note
state.json	38,882 B	
kicad/PCB-0007-A_rf-de-20m.kicad_pcb	889,703 B	
kicad/PCB-0007-A_rf-de-20m.kicad_sch	6,885 B	
fab/PCB-0007-A_rf-de-20m_gerbers.zip	165,232 B	
fab/BOM.csv	1,419 B	
fab/CPL.csv	2,014 B	
requirements.md	16,663 B	
architecture/blocks.md	37,342 B	
architecture/constraints.json	24,610 B	
architecture/decisions.md	30,793 B	
architecture/power_tree.md	17,822 B	
architecture/sheets.md	14,529 B	
architecture/stackup.md	18,766 B	
reports/bom_cpl.json	31,799 B	
reports/dfm_check-shipped.json	598 B	
reports/erc.json	261 B	
reports/fab_export.json	2,526 B	
reports/filter_dnp.json	3,181 B	
reports/gate-dfm.json	313 B	
reports/gate-drc_routed.json	40,084 B	
reports/gate-erc.json	373 B	
reports/gate-place.json	374 B	
reports/gate-sim.json	1,648 B	
reports/gate-verify.json	37,120 B	
reports/netlist_audit.json	1,489 B	
reports/plane_repair.json	7,126 B	
reports/review-board.json	10,477 B	
reports/review-board.md	27,499 B	
reports/review-schematic.json	7,817 B	
reports/review-schematic.md	20,810 B	
reports/route-notes.md	37,058 B	
reports/schematic.pdf	232,266 B	
reports/sim-notes.md	28,481 B	
reports/sim.json	9,967 B	
reports/spiral-design.md	16,842 B	
reports/top.net	122,342 B	

File	Size	Note
reports/verify-waivers.json	13,339 B	
reports/verify-waivers.md	25,635 B	
reports/verify.json	156,567 B	